



Australian Government

Department of Climate Change, Energy,
the Environment and Water

FullCAM PR External API Documentation

FullCAM Team

V1.2



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Table of Contents

FullCAM PR External API Documentation	1
Table of Contents	3
High-Level Overview of the Document	10
Request for Access to APIs:.....	12
Access Request Template:	12
Response	12
API Architecture.....	12
FullCAM 2020 Data Builder API	13
Overview	13
Key Features	13
API Authentication.....	13
Base URL	13
API Endpoints.....	14
1. siteinfo API	14
Parameters	14
Example Request	15
Example Response	15
2. species API.....	16
Parameters	16
Example Request	17
Example Response	17
3. regimes API	18
Parameters	18
Example Request	18
Example Response	18
4. templates API	19
Example Request	19
Example Response	19
5. template API.....	20
Parameters	20
Example Request	20
Example Response	20
6. update-spatialdata API.....	21
Request Body	21

Example Request	21
Example Response	22
Error Handling.....	22
Development Best Practices	24
1. Model Objects.....	24
2. Asynchronous Calls	24
3. Error Handling Mechanisms.....	24
4. Caching Strategy	24
5. Version Compatibility	24
Performance Optimization	25
Support and Contact	25
Changelog (2020)	26
FullCAM 2024 Data Builder API	27
Overview	27
Key Features.....	27
API Authentication.....	27
Base URL	27
1. siteinfo API	28
Parameters	28
Example Request	29
Example Response	29
2. species API.....	30
Parameters	30
Example Request	31
Example Response	31
3. regimes API	31
Parameters	31
Example Request	32
Example Response	32
4. templates API	32
Example Request	32
Example Response	33
5. template API.....	33
Parameters	33
Example Request	34

Example Response	34
6. convert-plotfile API	34
Request Body	34
Example Request	35
Example Response	35
7. update-spatialdata	36
Request Body	36
Example Request	36
Example Response	36
Error Handling	37
Error Response Format.....	37
Common Error Codes	37
Parameter Validation Errors	38
Development Best Practices	38
1. Model Objects.....	38
2. Asynchronous Calls	38
3. Error Handling Mechanisms	39
4. Caching Strategy	39
5. Rate Limiting Considerations	39
API Usage Workflow.....	39
Performance Optimization	40
Version Differences: 2020 vs 2024.....	40
Migration Considerations.....	41
FullCAM 2020 PR Plot Simulation API	42
Overview	42
Key Features.....	42
Authentication	42
Required Header	42
Base URL	42
1. run-plotsimulation API	43
Parameters	43
Request Headers	43
Example Request	43
Examples Response.....	43
2. run-pldfile-simulation API	44

Parameters	44
Request Headers	44
Example Request	45
Examples Response	45
3. run-est-filesimulation API.....	46
Parameters	46
Request Headers	46
Example Request	46
Response Examples	47
Development Best Practices	48
1. File Upload Considerations.....	48
2. Asynchronous Processing.....	48
3. Version Compatibility	49
Performance Optimization	49
FullCAM 2024 PR Plot Simulation API	49
Overview	49
Authentication	50
Base URL	50
1. run-plotsimulation API	50
Parameters	50
Request Headers	50
Example Request	51
Example Response	51
2. run-est-filesimulation API	52
Parameters	52
Request Headers	52
Example Request	52
Example Response	53
3. run-pld-filesimulation API	54
Parameters	54
Request Headers	54
Example Request	54
Example Response	55
Development Best Practices	56
1. File Upload Considerations.....	56

2. Enhanced Processing	56
3. Version Compatibility	56
Performance Optimization	57
API Usage Workflow	57
FullCAM Simulation Workflow API	59
Overview	59
Authentication	60
Base URL	60
Quick Start	59
Prerequisites	60
1. Upload Files to New Batch	61
Parameters	61
Request Header	61
Example Request	61
Example Response	62
2. Run Workflow for Batch	64
Parameters	65
Request Header	65
Request Body	65
Example Request	66
Example Response:	67
3. Check Simulation Status	68
Parameters	69
Request Header	69
Example Request	69
Example Response	69
4. Download Simulation Results	73
Parameter	73
Request Header	73
Example Request	73
Example Response	74
Data Models	75
Batch Identifier Format	75
PlotFileUploadResult	75
CreateBatchResponse	76

RunBatchRequest.....	76
StatusResponse.....	76
Error Handling.....	76
Error Response Format.....	76
Validation Error Format	77
Common Error Scenarios	77
Best Practices	78
Limits and Constraints	78
File Preparation.....	78
Upload Strategy	78
Monitoring Best Practices.....	78
Download Handling	78
Error Handling.....	79
Troubleshooting	79
Upload Issues.....	79
Workflow Issues.....	79
Download Issues	80
Performance Issues	80
RMT External API	81
Overview.....	81
Authentication	81
Base URL	81
1. RMT Site Info API:	81
Parameters	82
Request Headers	83
Example Request.....	83
Example Response	83
2. RMT Location Species Regimes API	85
Parameters	85
Request Headers	86
Example Request.....	87
Example Response	87
3. RMT Regime Details API	90
Parameters	90
Request Header.....	91

Example Request	91
Example Response	92
4. RMT Simulate API	96
Request Header	96
Request Body (form-data)	96
Example Request	97
Example Responses	97
5. RMT Decrypt File API.....	100
Request Headers	100
Request Body (Form Data).....	100
Example Requests	101
Support and Contact	104
API Usage Guide:	104
Glossary.....	104

High-Level Overview of the Document

This document provides a comprehensive guide to the FullCAM Modernisation APIs, covering both data-builder and simulation workflows across the FullCAM 2020, FullCAM 2024 Reforestation Modelling Tool (RMT). It explains how users can retrieve spatial, species, regime, and template data, update or convert plot files, and run simulations through plot, estate, digest, and batch-processing APIs. Together, these sections offer a full end-to-end view of how developers can programmatically access carbon-accounting datasets and automate modelling tasks using the FullCAM ecosystem.

This is a working document, and may be updated from time-to-time based on feedback received. To ensure you have the most recent version, check that the version number of the document matches the version number of the document that is downloadable from the Department's website. Please send any feedback on the document to fullcam@dcceew.gov.au.

API Groups Covered in this document

Below is a high-level summary of all API categories included:

FullCAM 2020 Data Builder APIs

- Get Site information
- Get Species information
- Get Regime information
- Get Templates and template details
- Update Spatial data for 2020 plot files

FullCAM 2024 Data Builder APIs

- Get Site information
- Get Species information
- Get Regime information
- Get Templates and template details
- Convert Plot file (2020 → 2024)
- Update Spatial data for 2020 and 2024 plot files

FullCAM 2020 Simulation APIs

- Run Plot simulation
- Run Plot digest simulation
- Run Estate file simulation

FullCAM 2024 Simulation APIs

- Run Plot simulation
- Run Plot digest simulation
- Run Estate file simulation

FullCAM Batch Simulation Workflow APIs

- Upload Plot file to create a Batch Job
- Trigger a Simulation workflow execution for the Batch job
- Check Workflow status
- Download the Batch Job Results

Reforestation Modelling Tool (RMT) APIs

Get Site Information
Get Location Species Regimes
Get Regime Details
Run RMT Simulation
Decrypt Legacy RMD File

Request for Access to APIs:

To access the FullCAM API, you must request a subscription key from the FullCAM business team.

Access Request Template:

Send an email to **fullcam@dcceew.gov.au** with the following information:

1. **Organization Name**
2. **Email Address**
3. **Phone Number**
4. **Business Use Case:** Detailed description of intended usage
5. **Usage Details:**
 - o Number of requests per system intended
 - o Number of users
 - o Peak time of usage
 - o Any other relevant usage patterns
6. **Number of Users**
7. **Access Duration:** How long you need access

Response

Once approved, you will be issued a **unique API subscription key** that must be included in all API requests.

Note: A separate subscription key will be issued for each FullCAM product, ensuring that access is provisioned in accordance with the specific API version or service requested.

API Architecture

FullCAM implements a REST (Representational State Transfer) architecture that is stateless, meaning servers do not save client data between requests.

FullCAM 2020 Data Builder API

Overview

The **Full Carbon Accounting Model (FullCAM) 2020 Data Builder API** provides programmatic access to Australia's carbon accounting data for the forestry and land use sectors. This RESTful API enables developers to retrieve site-specific information, species data, management regimes, and templates for carbon accounting calculations using the 2020 parameter set and calibrations.

Key Features

- **Geographic Site Information:** Soil, climate, and regional data for any Australian location
- **Species Database:** Comprehensive forest and agriculture species data with FullCAM 2020
- **Management Regimes:** Detailed forest and agriculture management protocols and practices
- **Template System:** Pre-configured carbon accounting templates for 2020 calculations
- **Cross-Platform Compatibility:** Removes dependency on Windows-only desktop application
- **Legacy Support:** Maintains compatibility with existing 2020 FullCAM workflows

API Authentication

All API requests require a subscription key for authentication.

Required Header

Ocp-Api-Subscription-Key: {{YOUR_SUBSCRIPTION_KEY}}

Error Response (401)

```
{
  "statusCode": 401,
  "message": "Access denied due to missing subscription key. Make sure to include subscription key when making requests to an API."
}
```

Base URL

<https://api.dcceew.gov.au/climate/carbon-accounting/2020/data/v1>

API Endpoints

1. siteinfo API

Retrieve comprehensive site information for a specific geographic location including soil, climate, and regional data using 2020 parameters.

API Endpoint: GET /2020/data-builder/siteinfo

Parameters

Parameter	Type	Required	Description	Valid Values
latitude	number	Yes	Latitude coordinate	-90 to 90
longitude	number	Yes	Longitude coordinate	-180 to 180
area	string	No	Spatial averaging area	Cell, Hectare, OneKm, TwoKm, ThreeKm, FiveKm
plotT	string	No	Plot type for analysis	SoilF, CompF, SoilA, CompA, CompM
frCat	string	No	Forest category filter	null, MVG, Plantation, EnvMallee, ERF, ERFH
incGrowth	boolean	No	Include growth parameters	true, false

Additional Note on Latitude & Longitude Validity

Latitude and longitude coordinates must be provided using the Geographic Datum of Australia (GDA) coordinate system, not the Australian Map Grid (AMG) system.

Mainland Australia and Tasmania extend approximately across the following coordinate ranges:

- **Latitude:** -10.6° (Cape York, Queensland) to -43.6° (South-east Cape, Tasmania)
- **Longitude:** 113.2° (Steep Point, Western Australia) to 153.6° (Cape Byron, New South Wales)

Inputs outside these Australian geographic bounds will result in no data availability or error responses, as FullCAM datasets only cover Australian land areas.

Area Parameter Values

- **Cell:** No spatial averaging
- **Hectare:** 1 hectare averaging
- **OneKm:** 100 hectare averaging

- **TwoKm**: 400 hectare averaging
- **ThreeKm**: 900 hectare averaging
- **FiveKm**: 2500 hectare averaging

Plot Type Values

- **SoilF**: Forest soil analysis
- **CompF**: Forest system analysis
- **SoilA**: Agricultural soil analysis
- **CompA**: Agricultural system analysis
- **CompM**: Mixed (forest and agricultural) system analysis

Forest Category Values

- **null**: All categories
- **MVG**: Native Vegetation Groups
- **Plantation**: Plantation Species
- **EnvMallee**: Environmental and Mallees
- **ERF**: ERF Methods
- **ERFH**: ERF Methods-EMP specific calibrations

Example Request

GET /2020/data-builder/siteinfo?latitude=-27.35757265&longitude=140.3277883&area=ThreeKm&plotT=CompF&frCat=ERF&incGrowth=true&version=2020

Host: api.dcceew.gov.au

Ocp-Apim-Subscription-Key: YOUR_SUBSCRIPTION_KEY

Example Response

```
<DocFragment Version="5007">
```

```
  <SiteInfo latitude="-27.35757265" longitude="140.3277883" areaBL="ThreeKm" state="SA"
  npi="Blank" growthRegion="0" SA2="41141" sa2Name="Outback" mvgTreeld="10" />
```

```
  <LocnSoil initFracBiof="0.0" initFracBios="0.0" initFracDpma="0.0"
  initFracRpma="0.0769481897354" initFracHums="0.7504163146019"
  initFracInrt="0.1726354777813" initTotalC="9.6754886627197" soilCoverA="0.6">
```

```
    <!-- Additional soil and climate data using 2020 parameters -->
```

```
  <TimeSeries tInTS="initTSMD" tExtrapTS="AvgYr" tOriginTS="Calendar" yr0TS="1970"
  nYrsTS="49">
```

```
    <!-- Time series data for soil moisture deficit -->
```

```

</TimeSeries>

</LocnSoil>

<!-- Species list for location using 2020 data -->

<ItemList count="6" id="FrSpecies">

  <ItemInfo id="106" value="Mallee eucalypt species" content="TYFParameters" />

  <ItemInfo id="104" value="Native Species Regeneration >=500mm rainfall"
content="TYFParameters" />

  <ItemInfo id="105" value="Native Species Regeneration <500mm rainfall"
content="TYFParameters" />

  <ItemInfo id="35" value="Mixed species environmental planting" content="TYFParameters" />

  <ItemInfo id="16" value="Eucalyptus globulus" content="TYFParameters" />

  <ItemInfo id="44" value="Pinus radiata" content="TYFParameters" />

</ItemList>

</DocFragment>

```

2. species API

Retrieve detailed species information including carbon coefficients and growth parameters using 2020 calibrations.

API Endpoint: GET /2020/data-builder/species

Parameters

Parameter	Type	Required	Description	Valid Values
latitude	number	Yes	Latitude coordinate	-90 to 90
longitude	number	Yes	Longitude coordinate	-180 to 180
area	string	No	Spatial averaging area	Cell, Hectare, OneKm, TwoKm, ThreeKm, FiveKm
frCat	string	No	Forest category filter	null, MVG, Plantation, EnvMallee, ERF, ERFH
specId	string	Yes	Species ID	Use the id value from the ItemInfo element in the siteInfo API response

Additional Note on Latitude & Longitude Validity

Latitude and longitude coordinates must be provided using the Geographic Datum of Australia (GDA) coordinate system, not the Australian Map Grid (AMG) system.

Mainland Australia and Tasmania extend approximately across the following coordinate ranges:

- Latitude: -10.6° (Cape York, Queensland) to -43.6° (South-east Cape, Tasmania)
- Longitude: 113.2° (Steep Point, Western Australia) to 153.6° (Cape Byron, New South Wales)

Inputs outside these Australian geographic bounds will result in no data availability or error responses, as FullCAM datasets only cover Australian land areas.

Getting Species ID

Species IDs should be extracted from the Site Info API response:

```
<ItemInfo id="23" value="Eucalyptus Species" content="Meta" />
```

```
<ItemInfo id="53" value="Pine Species" content="Meta" />
```

Use the id attribute value (e.g., "23", "53") as the specId parameter.

Example Request

```
GET /2020/data-builder/species?latitude=-27.35757265&longitude=140.3277883&specId=1&frCat=ERF&version=2020
```

Host: api.dcceew.gov.au

Ocp-Apim-Subscription-Key: YOUR_SUBSCRIPTION_KEY

Example Response

```
<DocFragment Version="5007">
```

```
<SpeciesForest idSP="35" nmSP="Mixed species environmental planting" grthModeSP="Yield"
tAgelxSP="AvgAge" curEdit="false" idRegimeSP="0" mvgTreeld="82" pltnType="Other"
tSpecFrCat="EnvMallee" CFracStemF="0.5" CFracBranF="0.468" CFracBarkF="0.487"
CFracLeafF="0.529" CFracCortF="0.492" CFracFirtF="0.461">
```

```
<!-- Detailed carbon fraction and growth data using 2020 parameters -->
```

```
<YieldClass>
```

```
<!-- Yield class specific parameters for 2020 -->
```

```
</YieldClass>
```

```
</SpeciesForest>
```

```
</DocFragment>
```

3. regimes API

Retrieve regime information for specific species and locations using 2020 protocols.

API Endpoint: GET /2020/data-builder/regimes

Parameters

Parameter	Type	Required	Description	Valid Values
latitude	number	Yes	Latitude coordinate	-90 to 90
longitude	number	Yes	Longitude coordinate	-180 to 180
frCat	string	No	Forest category filter	null, MVG, Plantation, EnvMallee, ERF, ERFH
specId	string	Yes	Species ID	Use the id value from the ItemInfo element in the siteInfo API response

Additional Note on Latitude & Longitude Validity

Latitude and longitude coordinates must be provided using the Geographic Datum of Australia (GDA) coordinate system, not the Australian Map Grid (AMG) system.

Mainland Australia and Tasmania extend approximately across the following coordinate ranges:

- **Latitude: -10.6° (Cape York, Queensland) to -43.6° (South-east Cape, Tasmania)**
- **Longitude: 113.2° (Steep Point, Western Australia) to 153.6° (Cape Byron, New South Wales)**

Inputs outside these Australian geographic bounds will result in no data availability or error responses, as FullCAM datasets only cover Australian land areas.

Example Request

GET /2020/data-builder/regimes?latitude=-27.35757265&longitude=140.3277883&specId=1&frCat=ERF&version=2020

Host: api.dcceew.gov.au

Ocp-Apim-Subscription-Key: YOUR_SUBSCRIPTION_KEY

Example Response

<DocFragment Version="5008">

<RegimeList idSP="1">

```
<RegimeF idRG="58" tSuite="ERF" nmRG="Acacia mangium (ERF 2020)" idSpecies="1"
tSpecFrCat="ERF" rotLenRGF="Long" stateRGF="Any" npiRGF="Blank" yrStRGF="1970"
yrEnRGF="2100" prevLUseRGF="Pasture" sitePrepRGF="None" ppWeedCntRGF="None"
cultRGF="Spot" thinRGF="Thin0" pruneRGF="Prune0">
```

```
<RegimeEvent seq="1" ruleSet="=212,CPN(CWSW)=-212GP" tevREV="PlnF" idSP="1"
eventNmREV="Plant trees" />
```

```
<!-- Additional regime events using 2020 protocols -->
```

```
</RegimeF>
```

```
</RegimeList>
```

```
</DocFragment>
```

4. templates API

Retrieve available carbon accounting templates for 2020 calculations.

API Endpoint: GET /2020/data-builder/templates

Example Request

GET /2020/data-builder/templates

Host: api.dcceew.gov.au

Ocp-Api-Subscription-Key: YOUR_SUBSCRIPTION_KEY

Example Response

```
<DocFragment Version="5007">
```

```
<ItemList count="3" id="Templates">
```

```
<ItemInfo id="1" value="ERF\Environmental Plantings Method.plo" content="Meta">
```

```
<Meta nmME="Template - Environmental Plantings Method" savedByResearch="false"
savedByVersion="" lockTime="" lockId="" lockOnME="false">
```

```
<notesME></notesME>
```

```
</Meta>
```

```
</ItemInfo>
```

```
<ItemInfo id="2" value="ERF\HIR NFMR and Avoided Clearing Methods greater than
500mm.plo" content="Meta">
```

```
<Meta nmME="Template - HIR NFMR and Avoided Clearing Methods"
savedByResearch="false" savedByVersion="" lockTime="" lockId="" lockOnME="false">
```

```
<notesME></notesME>
```

```
</Meta>
```

```

</ItemInfo>

<ItemInfo id="3" value="ERF\Additional Template.plo" content="Meta">

  <!-- Template metadata for 2020 -->

</ItemInfo>

</ItemList>

</DocFragment>

```

5. template API

Retrieve a specific template by name using 2020 parameters.

API Endpoint: GET /2020/data-builder/template

Parameters

Parameter	Type	Required	Description	Valid Values
templateName	string	Yes	Template name	Use the value from the ItemInfo element in the templates API response

Getting Template Name

Template names should be extracted from the Templates API response:

```

<ItemInfo id="1" value="ERF\Environmental Plantings Method 2020.plo" content="Meta" />

<ItemInfo id="2" value="ERF\HIR NFMR and Avoided Clearing Methods 2020.plo"
content="Meta" />

```

Use the value attribute (e.g., "ERF\Environmental Plantings Method 2020.plo") as the templateName parameter.

Example Request

GET /2020/data-builder/template?templateName=ERF\Environmental Plantings Method 2020.plo&version=2020

Host: api.dcceew.gov.au

Ocp-Apiim-Subscription-Key: YOUR_SUBSCRIPTION_KEY

Example Response

```

<DocFragment Version="5008">

  <DocumentPlot FileType="FullCAM Plot" Version="5008" pagelxDO="1" tDiagram="-1">

    <Meta nmME="Environmental Plantings Method 2020" savedByVersion="8.20.04.1101"
lockTime="" lockId="" lockOnME="false">

```

```

    <notesME>2020 calibrated template for environmental plantings</notesME>

</Meta>

<Config tPlot="CompF" userSoilMnrl="true" userMulchF="false" userMulchA="false">

    <!-- Detailed template configuration using 2020 parameters -->

</Config>

<!-- Complete template data structure with 2020 calibrations -->

</DocumentPlot>

</DocFragment>

```

6. update-spatialdata API

Update spatial data in FullCAM 2020 plot files while maintaining 2020 parameter compatibility.

API Endpoint: POST /2020/data-builder/update-spatialdata

Request Body

Parameter	Type	Required	Description
file	binary	Yes	Plot file requiring spatial data updates using 2020 parameters

Description

Upload a FullCAM 2020 plot file that requires spatial data updates. The API will refresh the spatial data using 2020 parameter sets and calibrations while maintaining compatibility with FullCAM 2020 PR workflows.

Key functionality:

- **Refresh location-specific data** using 2020 spatial datasets
- **Update soil information** with 2020 calibrated values
- **Maintain 2020 parameter compatibility** throughout the update process
- **Preserve existing plot structure** while updating spatial components

Content Type

- **Request:** multipart/form-data
- **File Format:** a

Example Request

POST /2020/data-builder/update-spatialdata

Host: api.dcceew.gov.au

Ocp-Apim-Subscription-Key: YOUR_SUBSCRIPTION_KEY

Content-Type: multipart/form-data

--boundary

Content-Disposition: form-data; name="file"; filename="my-plot-file-2020.xml"

Content-Type: application/xml

[Plot file content requiring 2020 spatial data update]

--boundary--

Example Response

```
<DocFragment Version="5008">
```

```
  <DocumentPlot FileType="FullCAM Plot" Version="5008" pagelxDO="1" tDiagram="-1">
```

```
    <!-- Updated plot file with refreshed 2020 spatial data -->
```

```
    <Meta nmME="Updated 2020 Plot File" savedByVersion="8.20.01.0000" />
```

```
    <SiteInfo latitude="-27.35757265" longitude="140.3277883" areaBL="ThreeKm" state="QLD" />
```

```
    <LocnSoil initFracBiof="0.0" initFracBios="0.0" initTotalC="45.0667" />
```

```
    <!-- Updated spatial data using 2020 parameters -->
```

```
  </DocumentPlot>
```

```
</DocFragment>
```

Error Handling

HTTP Status Codes

The FullCAM API uses standard HTTP status codes. Note that successful requests return a 200 status response code, even if no resources match the query (in which case an empty result set will be returned).

Error Response Format

```
{
  "type": "https://tools.ietf.org/html/rfc7231#section-6.5.1",
  "title": "Error description",
  "status": 400,
  "detail": "Detailed error message",
  "traceId": "00-trace-id-here-00",
  "errors": {
```

```

    "parameter": ["Validation error message"]
  }
}

```

Common Error Codes

Status Code	Description	Common Causes
200	Success	Request completed successfully (may return empty results)
400	Bad Request	Invalid parameter values, malformed request
401	Unauthorized	Missing or invalid subscription key
404	Not Found	Invalid API Endpoint, missing required parameters
500	Internal Server Error	Invalid parameter combinations, server issues

Parameter Validation Errors

Invalid Coordinates

```

{
  "status": 500,
  "detail": "BadRequest - Invalid latitude & longitude details. longitude:200,latitude:-27.35757265"
}

```

Invalid Parameter Format

```

{
  "status": 400,
  "title": "One or more validation errors occurred.",
  "errors": {
    "latitude": ["The value 'invalid' is not valid."]
  }
}

```

Invalid Species ID

```

<DocFragment Version="5008">
  <ItemList count="1" id="Messages">
    <ItemInfo id="1" value="Invalid SpeciesId" />
  </ItemList>
</DocFragment>

```

Development Best Practices

1. Model Objects

Define model objects according to the API documentation to handle appropriate responses from API calls. Structure your application to parse the XML response format properly. Note that 2020 responses use Version="5007" in XML fragments.

2. Asynchronous Calls

To avoid poor user experience with large delays, call the API's asynchronously rather than synchronously. This is especially important given the size of some responses (templates can be 700KB+).

3. Error Handling Mechanisms

Implement robust error handling to detect and respond to different HTTP status codes:

- **200:** Success (may include empty results)
- **400:** Client-side parameter errors
- **401:** Authentication failures
- **404:** API Endpoint not found
- **500:** Server-side processing errors

4. Caching Strategy

To improve performance and reduce API calls:

- **Cache spatial data locally** for quick searches rather than requesting from server each time
- **Refresh cache daily** to ensure you have the latest data
- **Cache templates and species data** as they change infrequently for 2020 version
- Consider implementing cache invalidation strategies

5. Version Compatibility

When working with FullCAM 2020 API:

- **Always specify version=2020** in parameters
 - **Maintain 2020 parameter sets** when processing data
 - **Be aware of differences** in carbon coefficients compared to 2024 version
 - **Test compatibility** if migrating from 2020 to 2024 workflows
-

Performance Optimization

Response Data Sizes (2020 Version)

Be aware that API responses can be quite large:

- **Site Info:** ~38-40KB per location
- **Species Data:** ~260-270KB per species
- **Templates:** ~1.1KB per template list
- **Regimes:** ~1-2KB per species
- **Plot File Operations:** Variable size depending on input file complexity

File Upload Considerations (2020)

- **Update Spatial Data:** Refreshes spatial data using 2020 parameter sets
- **File Size Limits:** Test with your specific file sizes to determine limits
- **Timeout Settings:** Use 60+ second timeouts for file upload operations
- **Version Maintenance:** Ensures plot files maintain 2020 parameter compatibility

Caching Recommendations (2020)

- **Cache site information** for frequently used locations with 2020 parameters
- **Cache templates daily** as 2020 templates are stable
- **Cache species data** for commonly used species with 2020 calibrations
- **Implement version-specific caching** to avoid conflicts with 2024 data
- **Do not cache file upload results** as they are typically one-time operations

Request Optimization (2020)

- **Always specify version=2020** in requests to ensure correct parameter sets
- **Batch requests** for multiple species when possible
- **Use appropriate timeout values** (30+ seconds for data, 60+ seconds for file uploads)
- **Implement retry logic** with exponential backoff
- **Process file uploads sequentially** rather than in parallel
- **Monitor parameter compatibility** when migrating between 2020 and 2024 versions

Support and Contact

- **Email:** fullcam@dcceew.gov.au
- **Terms of Service:** [FullCAM Terms](#)
- **License Agreement:** [FullCAM License](#)

Changelog (2020)

Version 2020

- Legacy API maintaining compatibility with FullCAM 2020 PR desktop application
 - RESTful architecture with subscription-based authentication
 - Support for Australian-wide geographic coverage using 2020 spatial datasets
 - Comprehensive species and regime databases with 2020 calibrations
 - Template system for standardized calculations using 2020 parameters
 - **File Processing Capabilities:**
 - o Plot file validation for FullCAM 2020 PR compatibility
 - o Spatial data updates using 2020 parameter sets
 - o Maintains compatibility with existing 2020 workflows
 - Cross-platform compatibility for web and non-web applications
 - **Legacy Support:** Ensures continuity for existing 2020-based carbon accounting workflows
-

FullCAM 2024 Data Builder API

Overview

The **Full Carbon Accounting Model (FullCAM) 2024 Data Builder API** provides programmatic access to Australia's comprehensive carbon accounting data for forestry and land use sectors. This RESTful API enables developers to retrieve site-specific information, species data, management regimes, and templates for carbon accounting calculations across various web and non-web based platforms.

Key Features

- **Geographic Site Information:** Soil, climate, and regional data for any Australian location
- **Species Database:** Comprehensive forest and agriculture species data with carbon coefficients
- **Management Regimes:** Detailed forest and agriculture management protocols and practices
- **Template System:** Pre-configured carbon accounting templates
- **Cross-Platform Compatibility:** Removes dependency on Windows-only desktop application
- **Real-time Data:** Current carbon accounting parameters and coefficients

API Authentication

All API requests require a subscription key for authentication.

Required Header

Ocp-Apim-Subscription-Key: {{YOUR_SUBSCRIPTION_KEY}}

Error Response (401)

```
{
  "statusCode": 401,
  "message": "Access denied due to missing subscription key. Make sure to include subscription key when making requests to an API."
}
```

Base URL

<https://api.dcceew.gov.au/climate/carbon-accounting/2024/data/v1>

1. siteinfo API

Retrieve comprehensive site information for a specific geographic location including soil, climate, and regional data.

API Endpoint: GET /2024/data-builder/siteinfo

Parameters

Parameter	Type	Required	Description	Valid Values
latitude	number	Yes	Latitude coordinate	-90 to 90
longitude	number	Yes	Longitude coordinate	-180 to 180
area	string	No	Spatial averaging area	Cell, Hectare, OneKm, TwoKm, ThreeKm, FiveKm
plotT	string	No	Plot type for analysis	SoilF, CompF, SoilA, CompA, CompM
frCat	string	No	Forest category filter	null, MVG, Plantation, EnvMallee, ERF, ERFH
incGrowth	boolean	No	Include growth parameters	true, false

Additional Note on Latitude & Longitude Validity

Latitude and longitude coordinates must be provided using the Geographic Datum of Australia (GDA) coordinate system, not the Australian Map Grid (AMG) system.

Mainland Australia and Tasmania extend approximately across the following coordinate ranges:

- **Latitude:** -10.6° (Cape York, Queensland) to -43.6° (South-east Cape, Tasmania)
- **Longitude:** 113.2° (Steep Point, Western Australia) to 153.6° (Cape Byron, New South Wales)

Inputs outside these Australian geographic bounds will result in no data availability or error responses, as FullCAM datasets only cover Australian land areas.

Area Parameter Values

- **Cell:** No spatial averaging
- **Hectare:** 1 hectare averaging
- **OneKm:** 100 hectare averaging
- **TwoKm:** 400 hectare averaging
- **ThreeKm:** 900 hectare averaging
- **FiveKm:** 2500 hectare averaging

Plot Type Values

- **SoilF:** Forest soil analysis
- **CompF:** Forest system analysis
- **SoilA:** Agricultural soil analysis
- **CompA:** Agricultural system analysis
- **CompM:** Mixed (forest and agricultural) system analysis

Forest Category Values

- **null:** All categories
- **MVG:** Native Vegetation Groups
- **Plantation:** Plantation Species
- **EnvMallee:** Environmental and Mallees
- **ERF:** ERF Methods
- **ERFH:** ERF Methods-EMP specific calibrations

Example Request

GET /2024/data-builder/siteinfo?latitude=-33.8688&longitude=151.2093&area=Cell&plotT=CompF&frCat=All&incGrowth=false

Host: api.dcceew.gov.au

Ocp-Apim-Subscription-Key: YOUR_SUBSCRIPTION_KEY

Example Response

<DocFragment Version="5009">

<SiteInfo latitude="-33.8688" longitude="151.2093" areaBL="Cell" state="NSW" npi="Blank" growthRegion="2" SA2="11337" sa2Name="Sydney - Haymarket - The Rocks" mvgTreeId="3" />

<LocnSoil initFracBiof="0.0" initFracBios="0.0" initFracDpma="0.0" initFracRpma="0.1751229614019" initFracHums="0.5600624084473" initFracInrt="0.2648146152496" initTotalC="49.0667381286621" soilCoverA="0.6">

<!-- Additional soil and climate data -->

</LocnSoil>

<!-- Species list for location -->

<ItemList count="145" id="Species">

<ItemInfo id="1" value="Acacia mangium" content="Meta" />

<ItemInfo id="23" value="Eucalyptus Species" content="Meta" />

<ItemInfo id="53" value="Pine Species" content="Meta" />

<!-- More species entries -->

</ItemList>
</DocFragment>

2. species API

Retrieve detailed species information including carbon coefficients and growth parameters.

API Endpoint: GET /2024/data-builder/species

Parameters

Parameter	Type	Required	Description	Valid Values
latitude	number	Yes	Latitude coordinate	-90 to 90
longitude	number	Yes	Longitude coordinate	-180 to 180
area	string	No	Spatial averaging area	Cell, Hectare, OneKm, TwoKm, ThreeKm, FiveKm
frCat	string	No	Forest category filter	null, MVG, Plantation, EnvMallee, ERF, ERFH
specId	string	Yes	Species ID	Use the id value from the ItemInfo element in the siteInfo API response

Additional Note on Latitude & Longitude Validity

Latitude and longitude coordinates must be provided using the Geographic Datum of Australia (GDA) coordinate system, not the Australian Map Grid (AMG) system.

Mainland Australia and Tasmania extend approximately across the following coordinate ranges:

- **Latitude:** -10.6° (Cape York, Queensland) to -43.6° (South-east Cape, Tasmania)
- **Longitude:** 113.2° (Steep Point, Western Australia) to 153.6° (Cape Byron, New South Wales)

Inputs outside these Australian geographic bounds will result in no data availability or error responses, as FullCAM datasets only cover Australian land areas.

Getting Species ID

Species IDs should be extracted from the Site Info API response:

```
<ItemInfo id="23" value="Eucalyptus Species" content="Meta" />
```

```
<ItemInfo id="53" value="Pine Species" content="Meta" />
```

Use the id attribute value (e.g., "23", "53") as the specId parameter.

Example Request

GET /2024/data-builder/species?latitude=-33.8688&longitude=151.2093&specId=1&frCat=Plantation

Host: api.dcceew.gov.au

Ocp-Apim-Subscription-Key: YOUR_SUBSCRIPTION_KEY

Example Response

<DocFragment Version="5009">

<SpeciesForest idSP="1" nmSP="Acacia mangium" grthModeSP="Yield" tAgeIxSP="AvgAge" curEdit="false" idRegimeSP="1" mvgtreeId="97" pltnType="Hardwood" tSpecFrCat="Plantation" CFracStemF="0.5" CFracBranF="0.468" CFracBarkF="0.487" CFracLeafF="0.529" CFracCortF="0.492" CFracFirtF="0.461">

<!-- Detailed carbon fraction and growth data -->

<YieldClass>

<!-- Yield class specific parameters -->

</YieldClass>

</SpeciesForest>

</DocFragment>

3. regimes API

Retrieve management regime information for specific species and locations.

API Endpoint: GET /2024/data-builder/regimes

Parameters

Parameter	Type	Required	Description	Valid Values
latitude	number	Yes	Latitude coordinate	-90 to 90
longitude	number	Yes	Longitude coordinate	-180 to 180
frCat	string	No	Forest category filter	null, MVG, Plantation, EnvMallee, ERF, ERFH
specId	string	Yes	Species ID	Use the id value from the ItemInfo element in the siteInfo API response

Additional Note on Latitude & Longitude Validity

Latitude and longitude coordinates **must be provided using the Geographic Datum of Australia (GDA)** coordinate system, **not** the Australian Map Grid (AMG) system.

Mainland Australia and Tasmania extend approximately across the following coordinate ranges:

- **Latitude:** -10.6° (Cape York, Queensland) to -43.6° (South-east Cape, Tasmania)
- **Longitude:** 113.2° (Steep Point, Western Australia) to 153.6° (Cape Byron, New South Wales)

Inputs outside these Australian geographic bounds will result in no data availability or error responses, as FullCAM datasets only cover Australian land areas.

Example Request

GET /2024/data-builder/regimes?latitude=-33.8688&longitude=151.2093&specId=1&frCat=ERF

Host: api.dcceew.gov.au

Ocp-Apim-Subscription-Key: YOUR_SUBSCRIPTION_KEY

Example Response

```
<DocFragment Version="5009">
```

```
<RegimeList idSP="1">
```

```
<RegimeF idRG="59" tSuite="NPI" nmRG="Acacia mangium (ERF)" idSpecies="1"
tSpecFrCat="All" rotLenRGF="Long" stateRGF="Any" npIRGF="Blank" yrStRGF="1970"
yrEnRGF="2100" prevLUseRGF="Pasture" sitePrepRGF="None" ppWeedCntRGF="None"
cultRGF="Spot" thinRGF="Thin0" pruneRGF="Prune0">
```

```
<RegimeEvent seq="1" ruleSet="=212,CPN(CWSW)=-212GP" tevREV="PlnF" idSP="1"
eventNmREV="Plant trees" />
```

```
<!-- Additional regime events -->
```

```
</RegimeF>
```

```
</RegimeList>
```

```
</DocFragment>
```

4. templates API

Retrieve available carbon accounting templates.

API Endpoint: GET /2024/data-builder/templates

Example Request

GET /2024/data-builder/templates

Host: api.dcceew.gov.au

Ocp-Apim-Subscription-Key: YOUR_SUBSCRIPTION_KEY

Example Response

```
<DocFragment Version="5009">

  <ItemList count="3" id="Templates">

    <ItemInfo id="1" value="ERF\Environmental Plantings Method.plo" content="Meta">

      <Meta nmME="Environmental Plantings Method" savedByVersion="8.23.04.1101"
lockTime="" lockId="" lockOnME="false">

        <notesME></notesME>

      </Meta>

    </ItemInfo>

    <ItemInfo id="2" value="ERF\HIR NFMR and Avoided Clearing Methods greater than
500mm.plo" content="Meta">

      <Meta nmME="HIR NFMR and Avoided Clearing Methods" savedByVersion="8.23.04.1101"
lockTime="" lockId="" lockOnME="false">

        <notesME></notesME>

      </Meta>

    </ItemInfo>

    <ItemInfo id="3" value="ERF\Additional Template.plo" content="Meta">

      <!-- Template metadata -->

    </ItemInfo>

  </ItemList>

</DocFragment>
```

5. template API

Retrieve a specific template by name.

API Endpoint: GET /2024/data-builder/template

Parameters

Parameter	Type	Required	Description	Valid Values
templateName	string	Yes	Template name	Use the value from the ItemInfo element in the templates API response

Getting Template Name

Template names should be extracted from the Templates API response:

```
<ItemInfo id="1" value="ERF\Environmental Plantings Method.plo" content="Meta" />
```

```
<ItemInfo id="2" value="ERF\HIR NFMR and Avoided Clearing Methods greater than 500mm.plo" content="Meta" />
```

Use the value attribute (e.g., "ERF\HIR NFMR and Avoided Clearing Methods greater than 500mm.plo") as the templateName parameter.

Example Request

```
GET /2024/data-builder/template?templateName=ERF\HIR NFMR and Avoided Clearing Methods greater than 500mm.plo&version=2024
```

Host: api.dcceew.gov.au

Ocp-Apim-Subscription-Key: YOUR_SUBSCRIPTION_KEY

Example Response

```
<DocFragment Version="5009">

  <DocumentPlot FileType="FullCAM Plot" Version="5009" pagelxDO="1" tDiagram="-1">

    <Meta nmME="HIR NFMR and Avoided Clearing Methods" savedByVersion="8.23.04.1101"
lockTime="" lockId="" lockOnME="false">

      <notesME></notesME>

    </Meta>

    <Config tPlot="CompF" userSoilMnrl="true" userMulchF="false" userMulchA="false">

      <!-- Detailed template configuration -->

    </Config>

    <!-- Complete template data structure -->

  </DocumentPlot>

</DocFragment>
```

6. convert-plotfile API

Convert FullCAM plot files from 2020 PR version to 2024 PR version.

API Endpoint: POST /2024/data-builder/convert-plotfile

Request Body

Parameter	Type	Required	Description
file	binary	Yes	plot file for version conversion

Description

Upload a FullCAM plot file that needs to be updated from FullCAM 2020 PR to FullCAM 2024 PR. The API will process the uploaded XML file and return the converted version with updated parameters and structure compatible with FullCAM 2024.

Content Type

- **Request:** multipart/form-data
- **File Format:** Plot file (.plo)

Example Request

POST /2024/data-builder/convert-plotfile

Host: api.dcceew.gov.au

Ocp-Apim-Subscription-Key: YOUR_SUBSCRIPTION_KEY

Content-Type: multipart/form-data

--boundary

Content-Disposition: form-data; name="file"; filename="my-plot-2020.xml"

Content-Type: application/xml

[Plot file content from FullCAM 2020 PR]

--boundary--

Example Response

<DocFragment Version="5009">

<DocumentPlot FileType="FullCAM Plot" Version="5009" pageIdxDO="1" tDiagram="-1">

<!-- Converted FullCAM 2024 PR plot file content -->

<Meta nmME="Converted Plot File" savedByVersion="8.24.01.0000" />

<!-- Updated parameters and structure for 2024 PR -->

</DocumentPlot>

</DocFragment>

7. update-spatialdata

Update spatial data in FullCAM plot files and optionally upgrade from 2020 PR to 2024 PR version.

API Endpoint: POST /2024/data-builder/update-spatialdata

Request Body

Parameter	Type	Required	Description
file	binary	Yes	plot file requiring spatial data updates

Description

Upload a FullCAM plot file that requires spatial data updates. The API behavior depends on the input file version:

- **FullCAM 2020 PR plot files:** The API will generate a plot file upgraded to FullCAM 2024 PR version, including updated spatial data with current parameters and coefficients.
- **FullCAM 2024 PR plot files:** The API will generate a plot file with refreshed spatial data, updating coordinates, soil information, climate data, and other location-specific parameters to current values.

Content Type

- **Request:** multipart/form-data
- **File Format:** Plot file (.plo)

Example Request

POST /2024/data-builder/update-spatialdata

Host: api.dcceew.gov.au

Ocp-Apim-Subscription-Key: YOUR_SUBSCRIPTION_KEY

Content-Type: multipart/form-data

--boundary

Content-Disposition: form-data; name="file"; filename="my-plot-file.xml"

Content-Type: application/xml

[Plot file content requiring spatial data update]

--boundary--

Example Response

<DocFragment Version="5009">

<DocumentPlot FileType="FullCAM Plot" Version="5009" pageIdxDO="1" tDiagram="-1">

```

<!-- Updated plot file with refreshed spatial data -->

<Meta nmME="Updated Plot File" savedByVersion="8.24.01.0000" />

<SiteInfo latitude="-33.8688" longitude="151.2093" areaBL="Cell" state="NSW" />

<LocnSoil initFracBiof="0.0" initFracBios="0.0" initTotalC="49.0667" />

<!-- Updated spatial data and parameters -->

</DocumentPlot>

</DocFragment>

```

Error Handling

HTTP Status Codes

The FullCAM API uses standard HTTP status codes. Note that successful requests return a 200 status response code, even if no resources match the query (in which case an empty result set will be returned).

Error Response Format

```

{
  "type": "https://tools.ietf.org/html/rfc7231#section-6.5.1",
  "title": "Error description",
  "status": 400,
  "detail": "Detailed error message",
  "traceId": "00-trace-id-here-00",
  "errors": {
    "parameter": ["Validation error message"]
  }
}

```

Common Error Codes

Status Code	Description	Common Causes
200	Success	Request completed successfully (may return empty results)
400	Bad Request	Invalid parameter values, malformed request
401	Unauthorized	Missing or invalid subscription key
404	Not Found	Invalid API Endpoint, missing required parameters

Status Code	Description	Common Causes
500	Internal Server Error	Invalid parameter combinations, server issues

Parameter Validation Errors

Invalid Coordinates

```
{
  "status": 500,
  "detail": "BadRequest - Invalid latitude & longitude details. longitude:200,latitude:-33.8688"
}
```

Invalid Parameter Format

```
{
  "status": 400,
  "title": "One or more validation errors occurred.",
  "errors": {
    "latitude": ["The value 'invalid' is not valid."]
  }
}
```

Invalid Species ID

```
<DocFragment Version="5009">
  <ItemList count="1" id="Messages">
    <ItemInfo id="1" value="Invalid SpeciesId" />
  </ItemList>
</DocFragment>
```

Development Best Practices

1. Model Objects

Define model objects according to the API documentation to handle appropriate responses from API calls. Structure your application to parse the XML response format properly.

2. Asynchronous Calls

To avoid poor user experience with large delays, call the web services asynchronously rather than synchronously. This is especially important given the size of some responses (templates can be 700KB+).

3. Error Handling Mechanisms

Implement robust error handling to detect and respond to different HTTP status codes:

- **200:** Success (may include empty results)
- **400:** Client-side parameter errors
- **401:** Authentication failures
- **404:** API Endpoint not found
- **500:** Server-side processing errors

4. Caching Strategy

To improve performance and reduce API calls:

- **Cache spatial data locally** for quick searches rather than requesting from server each time
- **Refresh cache daily** to ensure you have the latest data
- **Cache templates and species data** as they change infrequently
- Consider implementing cache invalidation strategies

5. Rate Limiting Considerations

While not explicitly documented, implement reasonable usage patterns:

- Avoid excessive concurrent requests
- Implement retry logic with exponential backoff
- Monitor your usage against your requested quotas

API Usage Workflow

Typical Carbon Accounting Data Retrieval Workflow

1. **Get Site Information** → Extract available species IDs and location-specific data
2. **Get Species Data** → For each relevant species using extracted IDs
3. **Get Management Regimes** → For specific management scenarios
4. **Get Templates** → For standardized calculation frameworks

Plot File Processing Workflow

1. **Convert Legacy Files** → Use convert-plotfile to upgrade FullCAM 2020 PR files to 2024 PR
2. **Update Spatial Data** → Use update-spatialdata to refresh location data in existing plot files
3. **Template Integration** → Download templates and integrate with updated plot files

Performance Optimization

Response Data Sizes

Be aware that API responses can be quite large:

- **Site Info:** ~40-60KB per location
- **Species Data:** ~200-220KB per species
- **Templates:** ~750KB per template
- **Regimes:** ~1-2KB per species
- **Plot File Operations:** Variable size depending on input file complexity

File Upload Considerations

- **Convert Plot File:** Processing time depends on file complexity (typically 5-30 seconds)
- **Update Spatial Data:** May involve version upgrades, allow extra processing time
- **File Size Limits:** Test with your specific file sizes to determine limits
- **Timeout Settings:** Use 60+ second timeouts for file upload operations

Caching Recommendations

- **Cache site information** for frequently used locations
- **Cache templates daily** as they change infrequently
- **Cache species data** for commonly used species
- **Implement cache invalidation** on a daily schedule
- **Do not cache file upload results** as they are typically one-time operations

Request Optimization

- **Batch requests** for multiple species when possible
- **Use appropriate timeout values** (30+ seconds for data, 60+ seconds for file uploads)
- **Implement retry logic** with exponential backoff
- **Limit concurrent requests** to avoid overwhelming the API
- **Process file uploads sequentially** rather than in parallel

Version Differences: 2020 vs 2024

Key Differences to Consider

Aspect	FullCAM 2020	FullCAM 2024	Migration Impact
Base URL	/carbon-accounting/2020/	/carbon-accounting/2024/	URL structure change
XML Version	Version="5007"	Version="5009"	Response format difference
Subscription Key	2020-specific key required	2024-specific key required	Separate authentication
Carbon Coefficients	2020 calibrated values	Updated 2024 calibrations	Data value differences
Parameter Sets	2020 parameter standards	Enhanced 2024 parameters	Calculation differences
Template Names	Standard template names	Template names may differ	File compatibility
Species Data	2020 species calibrations	Updated species data	Carbon fraction differences
Response Size	~38KB site info	~43KB site info	Slightly more compact

Migration Considerations

When transitioning from FullCAM 2020 to 2024:

1. **Data Validation:** Expect differences in carbon coefficients and calculations
2. **File Compatibility:** Use conversion API Endpoints to migrate plot files
3. **Parameter Updates:** Review and update any hardcoded parameter values
4. **Testing:** Thoroughly test calculations with both versions for comparison
5. **Documentation:** Update any documentation that references specific parameter values

FullCAM 2020 PR Plot Simulation API

Overview

The **FullCAM 2020 PR Plot Simulation API** provides programmatic access to run carbon accounting simulations using the FullCAM 2020 PR engine. This RESTful API enables developers to execute plot simulations, plot digest simulations, and estate file simulations using the 2020 parameter set and calibrations.

Key Features

- **Plot Simulation:** Execute individual plot file simulations using 2020 parameters
- **Plot Digest Simulation:** Run Scenario based simulations using plot digest files with 2020 calibrations
- **Estate File Simulation:** Process estate files with multiple associated plot files using 2020 protocols
- **Cloud Storage Integration:** Optional cloud storage for simulation results
- **Email Notifications:** User email integration for simulation tracking

Authentication

All API requests require a subscription key for authentication.

Required Header

Ocp-Apim-Subscription-Key: YOUR_2020_SUBSCRIPTION_KEY

Note: FullCAM 2020 and 2024 APIs require **separate subscription keys**:

- **FullCAM 2020:** Use subscription key for 2020 version access
- **FullCAM 2024:** Use subscription key for 2024 version access

Error Response (401)

```
{
  "statusCode": 401,
  "message": "Access denied due to missing subscription key. Make sure to include subscription key when making requests to an API."
}
```

Base URL

<https://api.climatechange.gov.au/climate/carbon-accounting/2020/plot/v1>

1. run-plotsimulation API

Execute a carbon accounting simulation using a single FullCAM plot file with 2020 parameters.

Endpoint: POST /2020/fullcam-simulator/run-plotsimulation

Parameters

Parameter	Type	Required	Location	Description
file	binary	Yes	Form Data	FullCAM plot file in .plo format
userEmail	string	No	Query	User email for simulation tracking

Request Headers

Header	Value	Required
Content-Type	multipart/form-data	Yes
Accept	text/plain	No
Ocp-Apim-Subscription-Key	YOUR_2020_SUBSCRIPTION_KEY	Yes

Example Request

POST /2020/fullcam-simulator/run-plotsimulation?userEmail=fullcam@dcceew.gov.au

Host: api.climatechange.gov.au

Content-Type: multipart/form-data

Accept: text/plain

Ocp-Apim-Subscription-Key: YOUR_2020_SUBSCRIPTION_KEY

--boundary

Content-Disposition: form-data; name="file"; filename="plot-2020.plo"

Content-Type: application/octet-stream

[Binary plot file content for FullCAM 2020 PR]

--boundary--

Examples Response

Success (200)

Simulation completed successfully

Bad Request (400)

```
{
  "type": "https://tools.ietf.org/html/rfc7231#section-6.5.1",
  "title": "Bad Request",
  "status": 400,
  "detail": "Invalid plot file format or missing required parameters",
  "instance": "/fullcam-simulator/run-plotsimulation"
}
```

Server Error (500)

Internal server error occurred during simulation processing

2. run-pldfile-simulation API

Execute batch simulations using a plot digest file with multiple plot configurations using 2020 parameters.

Endpoint: POST /2020/fullcam-simulator/run-pldfile-simulation

Parameters

Parameter	Type	Required	Location	Description	Valid Values
file	binary	Yes	Form Data	Plot digest file with outWinOption="Combined" in XML	
simOption	string	No	Form Data	Simulation output option	Combined, PerDoc, PerOutput

Request Headers

Header	Value	Required
Content-Type	multipart/form-data	Yes
Accept	text/plain	No
Ocp-Apim-Subscription-Key	YOUR_2020_SUBSCRIPTION_KEY	Yes

Simulation Options

- **Combined:** Generate a single combined output file
- **PerDoc:** Generate separate output files per document
- **PerOutput:** Generate separate files per output type

Example Request

POST /2020/fullcam-simulator/run-pldfile-simulation

Host: api.climatechange.gov.au

Content-Type: multipart/form-data

Accept: text/plain

Ocp-Apim-Subscription-Key: YOUR_2020_SUBSCRIPTION_KEY

--boundary

Content-Disposition: form-data; name="file"; filename="plotdigest-2020.pld"

Content-Type: application/xml

[Plot digest XML content with 2020 parameters]

--boundary

Content-Disposition: form-data; name="simOption"

Combined

--boundary--

Examples Response

Success (200)

Plot digest simulation completed successfully

Bad Request (400)

```
{
  "type": "https://tools.ietf.org/html/rfc7231#section-6.5.1",
  "title": "Bad Request",
  "status": 400,
  "detail": "Invalid plot digest file or simulation option",
  "instance": "/fullcam-simulator/run-pldfile-simulation"
```

}

Server Error (500)

Internal server error occurred during plot digest simulation

3. run-est-filesimulation API

Execute estate-level simulations using an estate file and associated plot files with 2020 protocols.

Endpoint: POST /2020/fullcam-simulator/run-est-filesimulation

Parameters

Parameter	Type	Required	Location	Description
estfile	binary	Yes	Form Data	Estate file in .est format
files	binary[]	Yes	Form Data	Array of plot files (.plo) referenced in estate file
userId	string	No	Form Data	User identifier for simulation tracking

Request Headers

Header	Value	Required
Content-Type	multipart/form-data	Yes
Accept	text/plain	No
Ocp-Apim-Subscription-Key	YOUR_2020_SUBSCRIPTION_KEY	Yes

Example Request

POST /2020/fullcam-simulator/run-est-filesimulation

Host: api.climatechange.gov.au

Content-Type: multipart/form-data

Accept: text/plain

Ocp-Apim-Subscription-Key: YOUR_2020_SUBSCRIPTION_KEY

--boundary

Content-Disposition: form-data; name="estfile"; filename="estate-2020.est"

Content-Type: application/octet-stream

[Estate file content for FullCAM 2020 PR]

--boundary

Content-Disposition: form-data; name="files"; filename="plot1-2020.plo"

Content-Type: application/octet-stream

[Plot file 1 content]

--boundary

Content-Disposition: form-data; name="files"; filename="plot2-2020.plo"

Content-Type: application/octet-stream

[Plot file 2 content]

--boundary

Content-Disposition: form-data; name="userId"

fullcam@dcceew.gov.au

--boundary--

Response Examples

Success (200)

Estate file simulation completed successfully

Bad Request (400)

```
{  
  "type": "https://tools.ietf.org/html/rfc7231#section-6.5.1",  
  "title": "Bad Request",  
  "status": 400,  
  "detail": "Invalid estate file or missing associated plot files",  
  "instance": "/fullcam-simulator/run-est-filesimulation"  
}
```

Server Error (500)

Internal server error occurred during estate file simulation

Error Handling

HTTP Status Codes

Status Code	Description	Common Causes
200	Success	Simulation completed successfully
400	Bad Request	Invalid file format, missing parameters, malformed request
401	Unauthorized	Missing or invalid subscription key
500	Internal Server Error	Simulation engine errors, file processing issues

Error Response Format

```
{  
  "type": "https://tools.ietf.org/html/rfc7231#section-6.5.1",  
  "title": "Error Title",  
  "status": 400,  
  "detail": "Detailed error description",  
  "instance": "/endpoint-path"  
}
```

Development Best Practices

1. File Upload Considerations

- **File Size Limits:** Test with your specific file sizes to determine limits
- **Timeout Settings:** Use 120+ second timeouts for simulation operations
- **File Format Validation:** Ensure plot files are valid FullCAM 2020 PR format
- **Estate File Dependencies:** Verify all referenced plot files are included in estate simulations

2. Asynchronous Processing

- Simulations may take significant time depending on complexity
- Implement proper timeout handling for long-running simulations

- Consider implementing polling mechanisms for simulation status

3. Version Compatibility

- **Use 2020 Subscription Key:** Ensure you're using the correct subscription key for 2020 version
- **File Version Validation:** Verify plot files are compatible with FullCAM 2020 PR
- **Parameter Sets:** Simulations use 2020 calibrated parameters and coefficients

Performance Optimization

Simulation Processing Times

Be aware that simulation times can vary significantly:

- **Single Plot Simulation:** 30 seconds to 5 minutes depending on complexity
- **Plot Digest Simulation:** 2-30 minutes depending on number of plots and complexity
- **Estate File Simulation:** 5-60 minutes depending on estate size and plot complexity

Request Optimization

- **Process simulations sequentially** rather than in parallel
- **Use appropriate timeout values** (120+ seconds for all simulation endpoints)
- **Implement retry logic** with exponential backoff for failed simulations
- **Monitor file sizes** to optimize upload and processing times

FullCAM 2024 PR Plot Simulation API

Overview

The **FullCAM 2024 PR Plot Simulation API** provides programmatic access to run carbon accounting simulations using the modernized FullCAM 2024 PR engine. This RESTful API enables developers to execute plot simulations, plot digest simulations, and estate file simulations using the latest 2024 parameter set and enhanced calibrations.

Key Features

- **Plot Simulation:** Execute individual plot file simulations using 2024 parameters
- **Estate File Simulation:** Process estate files with multiple associated plot files using 2024 protocols
- **Plot Digest Simulation:** Run batch simulations using plot digest files with 2024 calibrations
- **Enhanced Processing:** Improved simulation engine with faster processing times

- **Cloud Storage Options:** Configurable cloud storage for simulation results

Authentication

All API requests require a subscription key for authentication.

Required Header

Ocp-Apim-Subscription-Key: YOUR_2024_SUBSCRIPTION_KEY

Error Response (401)

```
{
  "statusCode": 401,
  "message": "Access denied due to missing subscription key. Make sure to include subscription key when making requests to an API."
}
```

Base URL

<https://api.climatechange.gov.au/climate/carbon-accounting/2024/plot/v1>

Endpoints

1. run-plotsimulation API

Execute a carbon accounting simulation using a single FullCAM plot file with 2024 parameters.

Endpoint: POST /2024/fullcam-simulator/run-plotsimulation

Parameters

Parameter	Type	Required	Location	Description
file	binary	Yes	Form Data	FullCAM plot file in .plo format

Request Headers

Header	Value	Required
Content-Type	multipart/form-data	Yes
Accept	text/plain	No
Ocp-Apim-Subscription-Key	YOUR_2024_SUBSCRIPTION_KEY	Yes

Example Request

POST /2024/fullcam-simulator/run-plotsimulation

Host: api.climatechange.gov.au

Content-Type: multipart/form-data

Accept: text/plain

Ocp-Apim-Subscription-Key: YOUR_2024_SUBSCRIPTION_KEY

--boundary

Content-Disposition: form-data; name="file"; filename="plot-2024.plo"

Content-Type: application/octet-stream

[Binary plot file content for FullCAM 2024 PR]

--boundary--

Example Response

Success (200)

Simulation completed successfully using FullCAM 2024 PR engine

Bad Request (400)

```
{
  "type": "https://tools.ietf.org/html/rfc7231#section-6.5.1",
  "title": "Bad Request",
  "status": 400,
  "detail": "Invalid plot file format or incompatible version",
  "instance": "/fullcam-simulator/run-plotsimulation"
}
```

Server Error (500)

Internal server error occurred during 2024 simulation processing

2. run-est-filesimulation API

Execute estate-level simulations using an estate file and associated plot files with 2024 protocols.

Endpoint: POST /2024/fullcam-simulator/run-est-filesimulation

Parameters

Parameter	Type	Required	Location	Description
estfile	binary	Yes	Form Data	Estate file in .est format
files	binary[]	Yes	Form Data	Array of plot files (.plo) referenced in estate file
userId	string	No	Form Data	User identifier for simulation tracking

Request Headers

Header	Value	Required
Content-Type	multipart/form-data	Yes
Accept	text/plain	No
Ocp-Apim-Subscription-Key	YOUR_2024_SUBSCRIPTION_KEY	Yes

Example Request

POST /2024/fullcam-simulator/run-est-filesimulation

Host: api.climatechange.gov.au

Content-Type: multipart/form-data

Accept: text/plain

Ocp-Apim-Subscription-Key: YOUR_2024_SUBSCRIPTION_KEY

--boundary

Content-Disposition: form-data; name="estfile"; filename="estate-2024.est"

Content-Type: application/octet-stream

[Estate file content for FullCAM 2024 PR]

--boundary

Content-Disposition: form-data; name="files"; filename="plot1-2024.plo"

Content-Type: application/octet-stream

[Plot file 1 content]

--boundary

Content-Disposition: form-data; name="files"; filename="plot2-2024.plo"

Content-Type: application/octet-stream

[Plot file 2 content]

--boundary

Content-Disposition: form-data; name="userId"

fullcam@dcceew.gov.au

--boundary--

Example Response

Success (200)

Estate file simulation completed successfully using FullCAM 2024 PR engine

Bad Request (400)

```
{
  "type": "https://tools.ietf.org/html/rfc7231#section-6.5.1",
  "title": "Bad Request",
  "status": 400,
  "detail": "Invalid estate file or missing associated plot files",
  "instance": "/fullcam-simulator/run-est-filesimulation"
}
```

Server Error (500)

Internal server error occurred during estate file simulation

3. run-pld-filesimulation API

Execute batch simulations using a plot digest file with multiple plot configurations using 2024 parameters.

Endpoint: POST /2024/fullcam-simulator/run-pld-filesimulation

Parameters

Parameter	Type	Required	Location	Description	Valid Values
file	binary	Yes	Form Data	Plot digest file with outWinOption="Combined" in XML	
simOption	string	No	Form Data	Simulation output option	Combined, PerDoc, PerOutput
saveOnCloud	boolean	No	Query	Save simulation results to cloud storage	true, false

Request Headers

Header	Value	Required
Content-Type	multipart/form-data	Yes
Accept	text/plain	No
Ocp-Apim-Subscription-Key	YOUR_2024_SUBSCRIPTION_KEY	Yes

Simulation Options

- **Combined:** Generate a single combined output file
- **PerDoc:** Generate separate output files per document
- **PerOutput:** Generate separate files per output type

Example Request

POST /2024/fullcam-simulator/run-pld-filesimulation?saveOnCloud=false

Host: api.climatechange.gov.au

Content-Type: multipart/form-data

Accept: text/plain

Ocp-Apim-Subscription-Key: YOUR_2024_SUBSCRIPTION_KEY

--boundary

Content-Disposition: form-data; name="file"; filename="plotdigest-2024.pld"

Content-Type: application/xml

[Plot digest XML content with 2024 parameters]

--boundary

Content-Disposition: form-data; name="simOption"

Combined

--boundary--

Example Response

Success (200)

Plot digest simulation completed successfully using FullCAM 2024 PR engine

Bad Request (400)

```
{
  "type": "https://tools.ietf.org/html/rfc7231#section-6.5.1",
  "title": "Bad Request",
  "status": 400,
  "detail": "Invalid plot digest file or simulation option",
  "instance": "/fullcam-simulator/run-pld-filesimulation"
}
```

Server Error (500)

Internal server error occurred during plot digest simulation

Error Handling

HTTP Status Codes

Status Code	Description	Common Causes
200	Success	Simulation completed successfully

Status Code	Description	Common Causes
400	Bad Request	Invalid file format, missing parameters, malformed request
401	Unauthorized	Missing or invalid subscription key
500	Internal Server Error	Simulation engine errors, file processing issues

Error Response Format

```
{
  "type": "https://tools.ietf.org/html/rfc7231#section-6.5.1",
  "title": "Error Title",
  "status": 400,
  "detail": "Detailed error description",
  "instance": "/endpoint-path"
}
```

Development Best Practices

1. File Upload Considerations

- **File Size Limits:** Test with your specific file sizes to determine limits
- **Timeout Settings:** Use 120+ second timeouts for simulation operations
- **File Format Validation:** Ensure plot files are valid FullCAM 2024 PR format
- **Version Compatibility:** Use plot files converted to 2024 PR format for optimal performance

2. Enhanced Processing

- FullCAM 2024 PR engine provides improved performance compared to 2020 version
- Better error handling and validation for simulation inputs
- Enhanced cloud storage integration options

3. Version Compatibility

- **Use 2024 Subscription Key:** Ensure you're using the correct subscription key for 2024 version
- **File Version Validation:** Verify plot files are compatible with FullCAM 2024 PR
- **Parameter Sets:** Simulations use latest 2024 calibrated parameters and coefficients

Performance Optimization

Simulation Processing Times (2024)

Enhanced performance with the 2024 PR engine:

- **Single Plot Simulation:** 20 seconds to 3 minutes depending on complexity (improved from 2020)
- **Plot Digest Simulation:** 1-20 minutes depending on number of plots and complexity (improved from 2020)
- **Estate File Simulation:** 3-40 minutes depending on estate size and plot complexity (improved from 2020)

Request Optimization

- **Process simulations sequentially** rather than in parallel
- **Use appropriate timeout values** (120+ seconds for all simulation endpoints)
- **Implement retry logic** with exponential backoff for failed simulations
- **Leverage cloud storage options** for result management
- **Use 2024 PR format files** for optimal processing performance

API Usage Workflow

Single Plot Simulation Workflow

1. **Prepare Plot File** → Ensure file is in FullCAM 2024 PR format (.plo)
2. **Submit Simulation** → Upload file via /run-plotsimulation endpoint
3. **Monitor Processing** → Wait for simulation completion (allow adequate timeout)
4. **Retrieve Results** → Process returned simulation output

Estate File Simulation Workflow

1. **Prepare Estate File** → Ensure estate file (.est) references are correct
2. **Gather Plot Files** → Collect all plot files referenced in estate file
3. **Submit Batch** → Upload estate file and all associated plot files
4. **Monitor Processing** → Wait for complete estate simulation
5. **Process Results** → Handle comprehensive estate-level output

Plot Digest Simulation Workflow

1. **Prepare Digest File** → Create plot digest (.pld) with outWinOption="Combined"
2. **Configure Options** → Select appropriate simulation options
3. **Submit Digest** → Upload digest file with simulation parameters

4. **Monitor Processing** → Wait for batch simulation completion
5. **Retrieve Results** → Process combined or individual simulation outputs

FullCAM Simulation Workflow API

Overview

The FullCAM Simulation Workflow API enables automated batch processing of simulations.

This API provides a straightforward four-step workflow:

1. Upload your plot files and receive a batch identifier
2. Start the simulation workflow
3. Monitor progress until completion
4. Download your results as a ZIP archive

Key Features

- Batch processing of multiple plot files in a single request
- Support for FullCAM 2020, FullCAM 2024, and RMT (Reforestation Modelling Tool) file formats
- Optional spatial data refresh for improved accuracy
- Email notifications upon completion
- Detailed per-file upload status reporting
- Comprehensive workflow state tracking

Quick Start

Here's the typical integration flow:

Step 1: Upload your plot files

POST /upload-files-to-new-batch

→ Returns: batchId

Step 2: Start the simulation

POST /run-workflow-for-batch

Body: { batchId, version, notificationEmail, ... }

→ Returns: 202 Accepted

Step 3: Poll for status (every 30-60 seconds)

GET /plot-simulation-status/{batchId}

→ Returns: { runtimeStatus, customStatus, lastUpdatedTime }

Step 4: Download results (when runtimeStatus = "Completed")

GET /simulation-result-package/{batchId}

→ Returns: ZIP file

Prerequisites

Before you begin, ensure you have:

1. A valid API subscription key from the Azure API Management portal
2. Input files in a supported format (FullCAM 2020, FullCAM 2024, or RMT)
3. A valid email address for receiving completion notifications

Authentication

All API requests require a subscription key for authentication.

Required Header

Ocp-Apim-Subscription-Key: YOUR_2024_SUBSCRIPTION_KEY

Error Response (401)

```
{  
  "statusCode": 401,  
  "message": "Access denied due to missing subscription key. Make sure to include subscription  
key when making requests to an API."  
}
```

Base URL

<https://api.dcceew.gov.au/climate/carbon-accounting/plotsimworkflow/v1>

1. Upload Files to New Batch

Create a new simulation batch and uploads your plot files. This endpoint stores your files and returns a batch identifier for use in subsequent operations. No processing begins at this stage.

Endpoint: POST /fullcam-simulator/batches

Parameters

Parameter	Type	Required	Description
plotFiles	File	Yes	One or more plot files to upload

Request Header

Header	Value	Required
Content-Type	multipart/form-data	Yes
Ocp-Apim-Subscription-Key	Your API key	Yes

Supported File Formats:

- FullCAM 2020 plot files (.plo)
- FullCAM 2024 plot files (.plo)
- RMT files (.rmd)

Constraints:

- At least one file required
- Maximum files: 1,024
- Maximum request size: 2,147,483,648 bytes (~2GB)

Example Request

POST /fullcam-simulator/batches

Host: {{baseUrlSimWrkFlw}}

Content-Type: multipart/form-data

Ocp-Apim-Subscription-Key: YOUR_SUBSCRIPTION_KEY

--boundary

Content-Disposition: form-data; name="plotFiles"; filename="project-2024.plo"

Content-Type: application/octet-stream

[Plot file content]

--boundary--

Example Response

Success Response (200 OK)

All files uploaded successfully:

```
{
  "batchId": "ext-20260130-142536-789-a1b2c3d4e5f67890abcdef1234567890",
  "plotFileUploadResults": [
    {
      "originalFileName": "ForestPlot_Site1.plo",
      "uploadedFileName": "ForestPlot_Site1.plo",
      "httpStatusCode": 201
    },
    {
      "originalFileName": "ForestPlot_Site2.plo",
      "uploadedFileName": "ForestPlot_Site2.plo",
      "httpStatusCode": 201
    },
    {
      "originalFileName": "AgriculturePlot.plo",
      "uploadedFileName": "AgriculturePlot.plo",
      "httpStatusCode": 201
    }
  ]
}
```

Partial Success Response (207 Multi-Status)

Some files uploaded, others failed:

```
{
  "batchId": "ext-20260130-142536-789-a1b2c3d4e5f67890abcdef1234567890",
  "plotFileUploadResults": [
    {
      "originalFileName": "ForestPlot_Site1.plo",
      "uploadedFileName": "ForestPlot_Site1.plo",
      "httpStatusCode": 201
    },
    {
      "originalFileName": "CorruptedFile.plo",
      "uploadedFileName": "CorruptedFile.plo",
      "httpStatusCode": 500
    },
    {
      "originalFileName": "CancelledUpload.plo",
      "uploadedFileName": "CancelledUpload.plo",
      "httpStatusCode": 499
    }
  ]
}
```

Error Response (400 Bad Request):

No files provided:

```
{
  "type": "https://tools.ietf.org/html/rfc7231#section-6.5.1",
  "title": "Bad Request",
  "status": 400,
  "detail": "No plot Files provided",
  "traceId": "00-abc123def456..."
}
```

Form value limit exceeded:

```
{
```

```

"type": "https://tools.ietf.org/html/rfc9110#section-15.5.1",
"title": "One or more validation errors occurred.",
"status": 400,
"errors": {
  "": [
    "Failed to read the request form. Form value count limit 1024 exceeded."
  ]
},
"traceId": "00-67733ebf4c97716fd0ab237784903985-b5f3729821ea3e91-00"
}

```

Request size limit exceeded:

```

{
  "type": "https://tools.ietf.org/html/rfc9110#section-15.5.1",
  "title": "One or more validation errors occurred.",
  "status": 400,
  "errors": {
    "": [
      "Failed to read the request form. Multipart body length limit 2147483648 exceeded."
    ]
  },
  "traceId": "00-d18c86aca440002cf8e56fb5d8c80e12-d80cedbf6a783103-00"
}

```

Per-File Status Codes:

Code	Status	Description
201	Created	File uploaded successfully
499	Client Closed Request	Upload was cancelled
500	Internal Server Error	Upload failed due to server error

2. Run Workflow for Batch

Initiates the simulation processing workflow for a previously uploaded batch. This starts the complete processing pipeline including optional spatial data updates, simulation execution, results generation, and email notification.

Endpoint: POST /fullcam-simulator/batches-run

Parameters

Parameter	Type	Required	Description
batchId	string	Yes	The batch identifier from the upload step
batchName	string	No	A friendly name for tracking and reference
version	string	Yes	FullCAM version: "2020", "2024", or "RMT"
notificationEmail	string	Yes	Email address for completion notification
isUpdatingSpatialAndSpecies	boolean	Yes	Whether to refresh spatial/species data

Request Header

Header	Value	Required
Content-Type	application/json	Yes
Ocp-Apim-Subscription-Key	Your API key	Yes

Request Body

```
{
  "batchId": "ext-20260130-142536-789-a1b2c3d4e5f67890abcdef1234567890",
  "batchName": "Q1 2026 Carbon Assessment - Forestry Project Alpha",
  "version": "2024",
  "notificationEmail": "carbon.analyst@example.gov.au",
  "isUpdatingSpatialAndSpecies": true
}
```

Version Parameter Details:

The `version` parameter must match your uploaded file format. This is critical for correct simulation results.

Value	Description	Use Case
-------	-------------	----------

"2020"	FullCAM 2020 format	Legacy plot files created with FullCAM 2020
"2024"	FullCAM 2024 format	Current plot files created with FullCAM 2024
"RMT"	Regional Mitigation Tool	RMT-specific analysis files

Spatial Data Update Option:

Value	Behaviour	Performance
true	Fetches latest spatial and species data from source systems	Slower (adds 5-10 minutes) but more accurate
false	Uses spatial and species data existing in the plot file	Faster processing

Limitations of isUpdatingSpatialAndSpecies flag:

Enabling this option will update spatial data based on the coordinates in the plot files, including climate information and species data.

- The **Update Spatial Data** feature is **not available** for **RMD files**.
- The **Update Spatial Data** feature works only for species of **the ACCU scheme**.
- **Initial Conditions** and **Events** pages will **not be updated** when using this feature.
- If a 2020 plot file is uploaded into the **2024 Batch Processing System**, species data will only be updated for certain **ACCU scheme species** (excluding plantation species).
- Updating spatial data **does not** update initial conditions or events data.

Example Request

POST /fullcam-simulator/batches-run

Host: {{baseUrlSimWrkFlw}}

Content-Type: application/json

Ocp-Apim-Subscription-Key: YOUR_SUBSCRIPTION_KEY

```
{
  "batchId": "ext-20240115-103045-123-a1b2c3d4e5f6",
  "batchName": "Q1 2024 Carbon Project",
  "version": "2024",
  "notificationEmail": "user@dcceew.gov.au",
  "isUpdatingSpatialAndSpecies": true
}
```

Example Response:

Status	Description
202 Accepted	Workflow started successfully
400 Bad Request	Invalid request or no files for batch
409 Conflict	Workflow already exists for this batch

Success Response (202 Accepted)

The response body is empty. A 202 status code indicates the workflow has been queued for processing.

Error Response (400 Bad Request)

No files exist for the specified batch:

```
{
  "type": "https://tools.ietf.org/html/rfc7231#section-6.5.1",
  "title": "Bad Request",
  "status": 400,
  "detail": "No plot Files exist for ext-20260130-142536-789-
a1b2c3d4e5f67890abcdef1234567890",
  "traceId": "00-abc123def456..."
}
```

Conflict Response (409 Conflict)

A workflow already exists for this batch:

```
{
  "message": "A workflow is already running. Please wait for it to complete. Batch: 'ext-
20260130-142536-789-...'",
  "batchId": "ext-20260130-142536-789-a1b2c3d4e5f67890abcdef1234567890",
  "status": "Running"
}
```

Other conflict messages:

```
{
```

```

"message": "A workflow is pending. Please wait for it to start. Batch: 'ext-...'",
"batchId": "ext-...",
"status": "Pending"
}
{
"message": "Workflow has already completed. Please create a new batch to run again. Batch:
'ext-...'",
"batchId": "ext-...",
"status": "Completed"
}

```

Workflow Conflict States

Current Status	Can Start New Workflow?	Resolution
Running	No	Wait for current workflow to complete
Pending	No	Wait for workflow to start processing
Completed	No	Create a new batch to run again
Failed	Yes	Can re-run the workflow
Terminated	Yes	Can re-run the workflow

Important Notes:

The `version` parameter must match your uploaded file format

Mismatched versions will cause simulation errors or incorrect results

Only one workflow can run per batch at a time

Failed or terminated workflows can be re-run with the same batch ID

3. Check Simulation Status

Retrieves the current processing status of a simulation workflow. Use this endpoint to monitor progress and determine when results are ready for download.

Endpoint: GET /fullcam-simulator/batches-status/{batchId}

Parameters

Path Parameters

Parameter	Type	Required	Description
batchId	string	Yes	The batch identifier

Query Parameters

Parameter	Type	Default	Description
includeDetails	boolean	false	Include full workflow execution details

Request Header

Header	Value	Required
Ocp-Apim-Subscription-Key	Your API key	Yes

Example Request

GET /fullcam-simulator/batches-status/ext-20240115-103045-123-a1b2c3d4e5f6?includeDetails=true

Ocp-Apim-Subscription-Key: YOUR_SUBSCRIPTION_KEY

Example Response

Status	Description
200 OK	Status retrieved successfully
404 Not Found	Workflow does not exist

Success Response (200 OK) - Running

Standard status response:

```
{
  "runtimeStatus": "Running",
  "customStatus": "RunPlotSimulationsCompleted",
  "lastUpdatedTime": "2026-01-30T14:35:00.0000000Z"
}
```

Success Response (200 OK) - Completed

```
{
  "runtimeStatus": "Completed",
```

```

"customStatus": "SendNotificationCompleted",
"lastUpdatedTime": "2026-01-30T14:55:00.0000000Z"
}

```

Success Response (200 OK) - Failed

```

{
  "runtimeStatus": "Failed",
  "customStatus": "Failed",
  "lastUpdatedTime": "2026-01-30T14:45:00.0000000Z"
}

```

Success Response (200 OK) - With Details

When `includeDetails=true`, returns the complete Durable Functions workflow status object with full execution history.

Error Response (404 Not Found)

```

"Workflow does not exist"

{
  "error": "Workflow instance not found",
  "batchId": "ext-20240115-103045-123-a1b2c3d4e5f6"
}

```

Runtime Status Values

The `runtimeStatus` field indicates the overall orchestration state (Azure Durable Functions):

Status	Description	Action Required
Pending	Workflow is queued, waiting to start	Continue polling
Running	Workflow is actively processing	Continue polling
Completed	All processing finished successfully	Download results
Failed	An error occurred during processing	Review error, possibly retry
Terminated	Workflow was manually stopped	Review and potentially restart

Custom Status Values

The `customStatus` field provides detailed progress within the workflow:

Status	Display Name	Description
Uninitiated	Not Started	Workflow initialisation pending
Initiated	Started	Workflow has begun execution
ReadBatchFileNamesCompleted	Reading Files	Input files have been identified
RunPlotSimulationsCompleted	Simulations Complete	All simulations have finished
CreateSimulationResultsPackageCompleted	Status Report Created	Results have been packaged
CreateBatchArchiveCompleted	Archive Created	ZIP file has been generated
SendNotificationCompleted	Notification Sent	Email notification has been sent
Failed	Failed	An error occurred

Response Examples by Stage

Just Started:

```
{
  "runtimeStatus": "Running",
  "customStatus": "Initiated",
  "lastUpdatedTime": "2026-01-30T14:30:00Z"
}
```

Reading Files:

```
{
  "runtimeStatus": "Running",
  "customStatus": "ReadBatchFileNamesCompleted",
  "lastUpdatedTime": "2026-01-30T14:31:00Z"
}
```

Running Simulations:

```
{
  "runtimeStatus": "Running",
  "customStatus": "RunPlotSimulationsCompleted",
  "lastUpdatedTime": "2026-01-30T14:45:00Z"
}
```

```
}
```

Creating Archive:

```
{  
  "runtimeStatus": "Running",  
  "customStatus": "CreateBatchArchiveCompleted",  
  "lastUpdatedTime": "2026-01-30T14:50:00Z"  
}
```

Completed:

```
{  
  "runtimeStatus": "Completed",  
  "customStatus": "SendNotificationCompleted",  
  "lastUpdatedTime": "2026-01-30T14:55:00Z"  
}
```

Polling Recommendations

Phase	Recommended Interval
Initial (first 2 minutes)	Every 10 seconds
Active processing	Every 30-60 seconds
Long-running (>15 minutes)	Every 60-120 seconds

Stop polling when `runtimeStatus` is `Completed`, `Failed`, or `Terminated`.

Expected Processing Times

Batch Size	Spatial Update	Estimated Duration
1-10 files	No	5-10 minutes
1-10 files	Yes	10-15 minutes
11-50 files	No	10-20 minutes
11-50 files	Yes	15-25 minutes
51-100 files	No	20-30 minutes
51-100 files	Yes	25-35 minutes

4. Download Simulation Results

Downloads the complete simulation results as a ZIP archive. This endpoint streams the file directly to your client.

Endpoint: GET /fullcam-simulator/simulation-result-package/{batchId}

Parameter

Parameter	Type	Required	Description
batchId	string	Yes	The batch identifier

Request Header

Header	Value	Required
Ocp-Apim-Subscription-Key	Your API key	Yes

Prerequisites

Before calling this endpoint, ensure:

- The workflow has completed (check `runtimeStatus` = "Completed")
- The archive has been created (check `customStatus` = "CreateBatchArchiveCompleted" or later)

Example Request

GET /fullcam-simulator/simulation-result-package /ext-20240115-103045-123-a1b2c3d4e5f6

Ocp-Apim-Subscription-Key: YOUR_SUBSCRIPTION_KEY

Response Codes

Status	Description
200 OK	File download starting
404 Not Found	Archive not found or not ready

Example Response

Success Response (200 OK)

The response is a binary file stream:

HTTP/1.1 200 OK

Content-Type: application/zip

Content-Disposition: attachment; filename="SimulationResults-ext-20260130-142536-789-a1b2c3d4e5f67890abcdef1234567890.zip"

[Binary ZIP file content]

Response Header:

Header	Example Value	Description
Content-Type	application/zip	MIME type indicating ZIP archive
Content-Disposition	attachment; filename="SimulationResults-{batchId}.zip"	Instructs browser to download

Error Response (404 Not Found)

```
{
  "message": "Archive for batch 'ext-20260130-142536-789-a1b2c3d4e5f67890abcdef1234567890' not found.",
  "batchId": "ext-20260130-142536-789-a1b2c3d4e5f67890abcdef1234567890"
}
```

ZIP Archive Contents

The downloaded archive contains:

File Type	Description
CSV Result Files	Carbon calculation results for each plot (` {PlotFileName}.csv`)
Original Plot Files	Your uploaded input files (preserved in original format)
Status Report	Summary of simulation success/failure per file

Error Logs	Detailed error information for failed simulations
Metadata	Batch ID, FullCAM version used, processing timestamp

File Naming Convention

SimulationResults-{batchId}.zip

Example: `SimulationResults-ext-20260130-142536-789-a1b2c3d4e5f67890abcdef1234567890.zip`

Data Models

Batch Identifier Format

ext-{yyyyMMdd}-{HHmmss}-{fff}-{guid32}

Component	Description	Example
ext	Prefix indicating external API	ext
yyyyMMdd	Date (UTC)	20260130
HHmmss	Time (UTC)	142536
fff	Milliseconds	789
guid32	32-character GUID (no hyphens)	a1b2c3d4e5f67890abcdef1234567890

Example: `ext-20260130-142536-789-a1b2c3d4e5f67890abcdef1234567890`

PlotFileUploadResult

```
{
  "originalFileName": "string",
  "uploadedFileName": "string",
  "httpStatusCode": 201
}
```

Field	Type	Description
originalFileName	string	The file name as submitted by client
uploadedFileName	string	The sanitised file name stored in system
httpStatusCode	integer	Upload result code (201, 499, 500)

CreateBatchResponse

```
{
  "batchId": "string",
  "plotFileUploadResults": [
    {
      "originalFileName": "string",
      "uploadedFileName": "string",
      "httpStatusCode": 201
    }
  ]
}
```

RunBatchRequest

```
{
  "batchId": "string",
  "batchName": "string",
  "version": "string",
  "notificationEmail": "string",
  "isUpdatingSpatialAndSpecies": true
}
```

StatusResponse

```
{
  "runtimeStatus": "string",
  "customStatus": "string",
  "lastUpdatedTime": "2026-01-30T14:35:00.0000000Z"
}
```

Error Handling

Error Response Format

Most API errors follow the RFC 7807 Problem Details format:

```
{
```

```

"type": "https://tools.ietf.org/html/rfc7231#section-6.5.1",
"title": "Bad Request",
"status": 400,
"detail": "Detailed description of the error",
"traceId": "00-abc123def456789..."
}

```

Validation Error Format

Validation errors include field-specific details:

```

{
  "type": "https://tools.ietf.org/html/rfc9110#section-15.5.1",
  "title": "One or more validation errors occurred.",
  "status": 400,
  "errors": {
    "fieldName": [
      "Error message 1",
      "Error message 2"
    ]
  },
  "traceId": "00-abc123def456789..."
}

```

Common Error Scenarios

Scenario	Status	Resolution
No files provided	400	Include at least one file in the upload
Form limit exceeded	400	Reduce number of files or use multiple batches
Request too large	400	Reduce total file size below 2GB
Batch not found	400/404	Verify batch ID is correct
Workflow conflict	409	Wait for existing workflow or create new batch
Archive not found	404	Ensure workflow completed before downloading

Authentication failed	401	Check API key is valid and included
-----------------------	-----	-------------------------------------

Best Practices

Limits and Constraints

Request Limits

Constraint	Limit	Notes
Maximum request size	2 GB	Total size of all files in upload
Maximum form values	1,024	Includes file metadata

File Preparation

1. Validate files locally before uploading to catch format issues early
2. Use consistent naming to easily identify files in results
3. Organise by project using the `batchName` parameter
4. Match versions carefully — ensure your files match the specified version parameter

Upload Strategy

1. Batch related files together for easier result correlation
2. Check individual file statuses when receiving a 207 response
3. Retry failed uploads by creating a new batch with only the failed files
4. Store batch IDs securely — they're needed for all subsequent operations

Monitoring Best Practices

1. Implement exponential backoff to reduce unnecessary API calls
2. Monitor both status fields (``runtimeStatus`` and ``customStatus``)
3. Set reasonable timeouts and alert if processing exceeds expected duration
4. Log all status responses for troubleshooting

Download Handling

1. Verify completion first — always check ``runtimeStatus`` before downloading
2. Implement retry logic for network failures during large downloads
3. Validate ZIP integrity after download completes

4. Store results appropriately based on data classification requirements

Error Handling

1. Parse error responses to extract actionable information
2. Preserve trace IDs for support escalations
3. Distinguish retrievable errors (5xx) from permanent errors (4xx)
4. Implement circuit breakers for resilience

Troubleshooting

Upload Issues

Problem: "No plot Files provided" error

Ensure files are attached with the parameter name `plotFiles`

Verify Content-Type is `multipart/form-data`

Check that at least one file is included

Problem: "Form value count limit 1024 exceeded"

Reduce the number of files per upload

Split into multiple batches if needed

Problem: "Multipart body length limit 2147483648 exceeded"

Total upload size exceeds 2GB

Reduce file sizes or split into multiple batches

Problem: 207 response with some files showing error codes

Check `statusCode` for each file in the response

Retry failed files in a separate batch

Investigate files with 500 errors for corruption

Workflow Issues

Problem: 409 Conflict when starting workflow

Check if a workflow is already running for this batch

Wait for completion or create a new batch

For "Completed" status, create a new batch to re-run

Problem: Workflow stuck in "Pending" state

System may be under heavy load

Continue polling — workflow should start within 15 minutes

Contact support if pending exceeds 30 minutes

Problem: Workflow status is "Failed"

Check if files are valid FullCAM format

Verify version parameter matches file format

Review error details in the response

Retry with ``isUpdatingSpatialAndSpecies: false`` to isolate the issue

Download Issues

Problem: 404 when downloading results

Verify workflow has completed (``runtimeStatus: "Completed"``)

Ensure ``customStatus`` has reached "CreateBatchArchiveCompleted"

Wait a few seconds after completion and retry

Verify batch ID is correct

Problem: Download interrupted or incomplete

Check network connectivity

Implement resume capability for large files

Retry the download request

Validate ZIP file integrity after completion

Performance Issues

Problem: Simulations taking longer than expected

Large batches naturally take longer

`isUpdatingSpatialAndSpecies: true` adds significant time (5-10 minutes)

System load during peak hours may affect performance

Contact support if processing exceeds 45 minutes

RMT External API

Overview

The Reforestation Modelling Tool (RMT) API provides comprehensive endpoints for carbon accounting and reforestation modelling in Australia. This API enables users to simulate reforestation scenarios, retrieve site-specific environmental data, query available species and management regimes, decrypt encrypted rmd files

Authentication

All API requests require a subscription key for authentication.

Required Header

Ocp-Apim-Subscription-Key: YOUR_rmt_SUBSCRIPTION_KEY

Error Response (401)

```
{
  "statusCode": 401,
  "message": "Access denied due to missing subscription key. Make sure to include subscription key when making requests to an API."
}
```

Base URL

<https://api.dcceew.gov.au/climate/carbon-accounting/reforestation-modelling/v1>

1. RMT Site Info API:

Purpose: Retrieve comprehensive site-specific environmental and ecological data for a given geographic location. This endpoint provides all necessary baseline information for reforestation modelling.

Endpoint Path: /rmt/site-info

Description

This endpoint returns detailed site information for a specific geographic coordinate including climate data (rainfall, temperature), forest productivity index, maximum above-ground biomass multiplication factor, location details (state, NPI region), initial species information with comprehensive carbon fraction and allocation parameters, and a list of all available species

suitable for the location. This data forms the foundation for accurate reforestation modelling and carbon accounting.

Parameters

Parameter Name	Type	Required	Description	Valid Values
lon	number	Yes	Longitude coordinate of the site location	Must be valid longitude value between -180 and 180
lat	number	Yes	Latitude coordinate of the site location	Must be valid latitude value between -90 and 90

Additional Note on Latitude & Longitude Validity

Latitude and longitude coordinates must be provided using the Geographic Datum of Australia (GDA) coordinate system, not the Australian Map Grid (AMG) system.

Mainland Australia and Tasmania extend approximately across the following coordinate ranges:

- **Latitude:** –10.6° (Cape York, Queensland) to –43.6° (South-east Cape, Tasmania)
- **Longitude:** 113.2° (Steep Point, Western Australia) to 153.6° (Cape Byron, New South Wales)

Inputs outside these Australian geographic bounds will result in no data availability or error responses, as RMT datasets only cover Australian land areas.

Query Parameter Validation Rules

1. **Longitude Range:** Must be between -180 and 180 (inclusive)
2. **Latitude Range:** Must be between -90 and 90 (inclusive)
3. **Precision:** Coordinates should have sufficient precision (typically 6 decimal places for accurate location)
4. **Australian Territory:** For RMT purposes, coordinates must fall within Australian territory
5. **Data Availability:** System must have data available for the specified location
6. **Format:** Must be numeric values (decimal format)

Request Headers

Header Name	Data Type	Value	Required	Description	Purpose
Accept	string	text/plain	Yes	Specifies the expected response format	Indicates client expects plain text response (though actual response is JSON)
Ocp-Apim-Subscription-Key	string	{subscription_key}	Yes	Azure APIM subscription key	Validates API subscription and tracks usage

Example Request

GET {{baseUrlrmt}}api/v1/rmt/site-info?lon=136.551911&lat=-11.385201

Accept: text/plain

Ocp-Apim-Subscription-Key: YOUR_rmt_SUBSCRIPTION_KEY

Example Response

Success Response (200 OK)

```
{
  "status": "integer",
  "data": {
    "Location": {
      "Longitude": "double",
      "Latitude": "double",
      "State": "string (enum)",
      "NpiRegion": "string (enum)"
    },
    "MaxAbgMf": "double",
    "ForestProdIx": {
      "Rainfall": {
        "AvgAirTemp": {
          "...TimeSeriesObject..."
        }
      }
    },
    "...TimeSeriesObject..."
  }
}
```

```

    },
    "...TimeSeriesObject..."
  },
  "InitialSpecies": {
    "...SpeciesObject..."
  },
  "AvailableSpecies": [
    "...SpeciesArray..."
  ]
},
"message": "string"
}

```

Troubleshooting

Problem: 400 Bad Request with "No data available"

Solution: Verify coordinates are within Australian territory. Some remote locations or offshore areas may lack data coverage. Try nearby coordinates or contact support for data availability.

Problem: Unexpected InitialSpecies returned

Solution: InitialSpecies is system-determined based on location and ecological suitability. Use AvailableSpecies list to select alternatives if needed.

Problem: Time series data interpretation errors

Solution: Check dataPerYrTS field - it indicates data frequency (1=annual, 12=monthly, etc.). Use multTS to scale raw values. For monthly data with 12 values, each represents one month.

Problem: Large response size causing timeouts or memory issues

Solution: Response includes extensive species data with many parameters. Ensure adequate timeout settings (recommend 30-60 seconds). Consider pagination if available or request specific fields only.

Problem: Inconsistent results for nearby coordinates

Solution: Small coordinate differences can cross regional boundaries or climate zones. This is expected behavior. Use exact same coordinates for consistency across related API calls.

Problem: Missing or null values in InitialSpecies parameters

Solution: Some parameters may not be available for all species or locations. Implement null-safe handling. Use default values where appropriate or contact support for missing data.

Problem: Cannot find species ID in AvailableSpecies list

Solution: Species availability varies by location. The species you're looking for may not be suitable for this location. Review the complete AvailableSpecies list for alternatives.

2. RMT Location Species Regimes API

Purpose: Retrieve available management regimes for a specific species at a given location. Management regimes define silvicultural practices including site preparation, weed control, and intended end product.

Endpoint Path: /rmt/location-species-regimes

Description

This endpoint returns a list of applicable management regimes for a specified species at a particular geographic location. Each regime represents a different combination of site preparation methods, post-planting weed control strategies, and intended end products (e.g., pulpwood, sawlog, firewood). This information is essential for accurate reforestation modelling as different management practices significantly impact carbon sequestration rates, biomass accumulation, and economic outcomes.

Parameters

Parameter Name	Type	Required	Description	Valid Values
lon	number	Yes	Longitude coordinate of the site location	Must be valid longitude value between -180 and 180
lat	number	Yes	Latitude coordinate of the site location	Must be valid latitude value between -90 and 90.
speciesId	string	Yes	Unique identifier for the species	Must be a valid species ID from the system. Obtain from site-info endpoint's AvailableSpecies list or InitialSpecies.Id.

Additional Note on Latitude & Longitude Validity

Latitude and longitude coordinates must be provided using the Geographic Datum of Australia (GDA) coordinate system, not the Australian Map Grid (AMG) system.

Mainland Australia and Tasmania extend approximately across the following coordinate ranges:

- **Latitude:** -10.6° (Cape York, Queensland) to -43.6° (South-east Cape, Tasmania)
- **Longitude:** 113.2° (Steep Point, Western Australia) to 153.6° (Cape Byron, New South Wales)

Inputs outside these Australian geographic bounds will result in no data availability or error responses, as RMT datasets only cover Australian land areas.

Query Parameter Validation Rules

1. **Longitude Range:** Must be between -180 and 180 (inclusive)
2. **Latitude Range:** Must be between -90 and 90 (inclusive)
3. **Species ID Format:** Must be a valid alphanumeric species identifier
4. **Species Availability:** Species must be available for the specified location (verify using site-info endpoint)
5. **Data Availability:** System must have regime data for the species-location combination
6. **Parameter Presence:** All three parameters (lon, lat, speciesId) are required

Request Headers

Header Name	Value	Required	Data Type	Description	Purpose
Accept	text/plain	Yes	string	Specifies the expected response format	Indicates client expects plain text response (though actual response is JSON)
Authorization	Bearer {token}	Yes	string	OAuth 2.0 bearer token for authentication	Authenticates the API request
Ocp-Apim-Subscription-Key	{subscription_key}	Yes	string	Azure APIM subscription key	Validates API subscription and tracks usage

Example Request

```
GET {{baseUrlrmt}}api/v1/rmt/location-species-regimes?lon=144&lat=-15&speciesId=c7db0a0134493d1019360000
```

Accept: text/plain

Ocp-Apim-Subscription-Key: YOUR_rmt_SUBSCRIPTION_KEY

Example Response

Success Response (200 OK)

```
{
  "status": "integer",
  "data": [
    {
      "Id": "integer",
      "Name": "string",
      "SitePrep": "string (enum)",
      "PostPlantingWeedControl": "string (enum)",
      "EndProduct": "string (enum)"
    }
  ],
  "message": "string"
}
```

Bad Request Response (400) - Invalid Species

```
{
  "type": "https://api.dcceew.gov.au/errors/invalid-species",
  "title": "Invalid Species",
  "status": 400,
  "detail": "The specified species ID 'invalid_species_id' is not valid or not available for the location (lon: 144, lat: -15). Please use a valid species ID from the site-info endpoint.",
  "instance": "/rmt/location-species-regimes"
}
```

Common 400 Error Scenarios:

- Invalid species ID format
- Species not available for specified location
- Longitude out of range (-180 to 180)
- Latitude out of range (-90 to 90)
- Missing required query parameters (lon, lat, or speciesId)
- Species ID not found in system
- No regimes available for species-location combination (returns empty array, not error)
- Invalid parameter data types (non-numeric lon/lat)

Unauthorized Response (401)

```
{  
  "type": "https://api.dcceew.gov.au/errors/unauthorized",  
  "title": "Unauthorized",  
  "status": 401,  
  "detail": "Authentication is required to access this endpoint. Please provide a valid  
    OAuth 2.0 Bearer token in the Authorization header.",  
  "instance": "/rmt/location-species-regimes"  
}
```

Common 401 Error Scenarios:

- Missing Authorization header
- Invalid or expired OAuth token
- Malformed Bearer token format
- Token revoked or invalidated

Server Error Response (500)

```
{  
  "type": "https://api.dcceew.gov.au/errors/internal-error",  
  "title": "Internal Server Error",  
  "status": 500,  
  "detail": "An unexpected error occurred while retrieving regime information. The  
    regime database service is temporarily unavailable. Please try again later or  
    contact support.",  
}
```



```
"instance": "/rmt/location-species-regimes"  
}
```

Common 500 Error Scenarios:

- Database connection failure
- Regime database query timeout
- Species-regime mapping service unavailable
- Memory issues with large regime datasets
- External service dependency failure

Troubleshooting

Problem: 400 Bad Request with "Species not available"

Solution: Verify species ID is from the AvailableSpecies list for this specific location from site-info endpoint. Species availability varies by location due to ecological suitability.

Problem: Empty data array returned (no regimes)

Solution: No regimes are defined for this species-location combination. This is valid response. Try a different species from the AvailableSpecies list or a different location.

Problem: Unexpected regime options returned

Solution: Regime availability depends on location-specific factors (climate, soil, regulations). This is expected behavior. Review regime details to understand suitability.

Problem: Regime IDs not matching documentation or expectations

Solution: Regime IDs are system-generated and may vary by location and species. Always use the ID returned by the API, not hardcoded values.

Problem: 401 Unauthorized error

Solution: This endpoint requires OAuth authentication. Ensure valid bearer token is included in Authorization header. Check token expiration and refresh if necessary.

Problem: Inconsistent results for same species at nearby locations

Solution: Regime availability can vary based on precise location due to regional factors. Small coordinate differences can cross boundaries. This is expected.

Problem: Cannot determine which regime to use

Solution: Use regime-details endpoint to get comprehensive information about each regime. Compare rotation lengths, management intensity, and expected outcomes.

3. RMT Regime Details API

Purpose: Retrieve comprehensive detailed information about a specific management regime at a given location. This endpoint provides all parameters needed to understand and implement a particular silvicultural regime.

Endpoint Path: /rmt/regime-details

Description

This endpoint returns detailed specifications for a specific management regime at a particular location. The response includes comprehensive information about the regime's characteristics, management practices, timing of operations, expected yields, and all parameters necessary for accurate carbon modelling. This data is essential for understanding the full implications of selecting a particular management regime for reforestation modelling and making informed decisions about plantation management strategies.

Parameters

Parameter Name	Type	Required	Description	Valid Values
lon	number	Yes	Longitude coordinate of the site location	Must be valid longitude value between -180 and 180.
lat	number	Yes	Latitude coordinate of the site location	Must be valid latitude value between -90 and 90.
regimeld	number	Yes	Unique identifier for the management regime	Must be a valid regime ID from the system. Obtain from location-species-regimes endpoint. Must be positive integer.

Additional Note on Latitude & Longitude Validity

Latitude and longitude coordinates must be provided using the Geographic Datum of Australia (GDA) coordinate system, not the Australian Map Grid (AMG) system.

Mainland Australia and Tasmania extend approximately across the following coordinate ranges:

- **Latitude:** -10.6° (Cape York, Queensland) to -43.6° (South-east Cape, Tasmania)
- **Longitude:** 113.2° (Steep Point, Western Australia) to 153.6° (Cape Byron, New South Wales)

Inputs outside these Australian geographic bounds will result in no data availability or error responses, as RMT datasets only cover Australian land areas.

Query Parameter Validation Rules

1. **Longitude Range:** Must be between -180 and 180 (inclusive)
2. **Latitude Range:** Must be between -90 and 90 (inclusive)
3. **Regime ID Format:** Must be a positive integer
4. **Regime Availability:** Regime must be available for the specified location
5. **Data Availability:** System must have detailed data for the regime-location combination
6. **Parameter Presence:** All three parameters (lon, lat, regimeId) are required

Request Header

Header Name	Value	Required	Data Type	Description	Purpose
Accept	text/plain	Yes	string	Specifies the expected response format	Indicates client expects plain text response (though actual response format may vary)
Authorization	Bearer {token}	Yes	string	OAuth 2.0 bearer token for authentication	Authenticates the API request
Ocp-Apim-Subscription-Key	{subscription_key}	Yes	string	Azure APIM subscription key	Validates API subscription and tracks usage

Example Request

GET {{baseUrlrmt}}api/v1/rmt/regime-details?lon=144&lat=-15®imeId=1050

Accept: text/plain

Authorization: Bearer eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...

Ocp-Apim-Subscription-Key: YOUR_rmt_SUBSCRIPTION_KEY.

Example Response

Success Response (200 OK)

The response structure for regime details is extensive and includes comprehensive information about the management regime. The exact structure may vary based on the regime type and available data.

Note: Based on the collection information, the success response may return an empty body or detailed regime information depending on data availability.

Expected Response Structure (based on typical regime detail requirements):

```
{
  "status": "integer",
  "data": {
    "RegimeId": "integer",
    "RegimeName": "string",
    "SpeciesId": "string",
    "SpeciesName": "string",
    "Location": {
      "Longitude": "double",
      "Latitude": "double",
      "State": "string",
      "NpiRegion": "string"
    },
    "ManagementPractices": {
      "SitePrep": "string",
      "PostPlantingWeedControl": "string",
      "EndProduct": "string",
      "PlantingDensity": "integer",
      "RotationLength": "integer"
    },
    "Operations": [
      {
        "OperationType": "string",
        "TimingYears": "double",
```

```

        "Description": "string"
    }
],
"YieldCurve": {
    "...TimeSeries data..."
},
"CarbonParameters": {
    "...Carbon accounting parameters..."
}
},
"message": "string"
}

```

Bad Request Response (400) - Invalid Regime

```

{
  "type": "https://api.dcceew.gov.au/errors/invalid-regime",
  "title": "Invalid Regime",
  "status": 400,
  "detail": "The specified regime ID '9999' is not valid or not available for the
    location (lon: 144, lat: -15). Please use a valid regime ID from the
    location-species-regimes endpoint.",
  "instance": "/rmt/regime-details"
}

```

Common 400 Error Scenarios:

- Invalid regime ID (non-existent or not applicable to location)
- Longitude out of range (-180 to 180)
- Latitude out of range (-90 to 90)
- Missing required query parameters (lon, lat, or regimeld)
- Regime not available for specified location
- Invalid parameter format (non-numeric values for lon/lat, non-integer for regimeld)
- Negative regime ID

Unauthorized Response (401)

```
{
  "type": "https://api.dcceew.gov.au/errors/unauthorized",
  "title": "Unauthorized",
  "status": 401,
}
```

Server Error Response (500)

```
{
  "type": "https://api.dcceew.gov.au/errors/internal-error",
  "title": "Internal Server Error",
  "status": 500,
  "detail": "An unexpected error occurred while retrieving regime details. The regime
    database service encountered an error. Please try again later or contact support.",
  "instance": "/rmt/regime-details"
}
```

Common 500 Error Scenarios:

- Database connection failure
- Regime detail database query timeout
- Complex calculation failure
- Memory issues with detailed regime data
- External service dependency failure

Troubleshooting

Problem: 400 Bad Request with "Invalid regime"

Solution: Verify regime ID is from the location-species-regimes endpoint for this specific location. Regime availability varies by location. Do not use hardcoded regime IDs.

Problem: Empty response body or null data

Solution: This may be expected behaviour if detailed data is not available for the regime. Use regime summary information from location-species-regimes endpoint. Contact support if detailed data is needed.

Problem: Regime details don't match expectations or documentation

Solution: Regime parameters are location-specific and may be adjusted based on site conditions (climate, soil, etc.). This is expected behavior for accurate modeling.

Problem: Missing expected fields in response

Solution: Response structure may vary by regime type and data availability. Not all regimes have all parameters. Implement null-safe handling for optional fields.

Problem: 401 Unauthorized error

Solution: This endpoint requires OAuth authentication. Ensure valid bearer token is included in Authorization header. Check token expiration and refresh if necessary.

Problem: Yield curve interpretation errors

Solution: Check dataPerYrTS and multTS fields in time series data. Apply multiplication factor (multTS) to raw values (rawTS) to get actual yield values. Understand time origin (tOriginTS).

Problem: Inconsistent results for same regime at different locations

Solution: Regime parameters are location-adjusted based on site-specific factors. This is expected and necessary for accurate modeling.

Problem: Cannot determine rotation length or harvest timing

Solution: Check Operations array for Harvest operation timing. If not available, use typical rotation lengths for the EndProduct type.

4. RMT Simulate API

Purpose: Execute a reforestation simulation based on uploaded configuration file. This endpoint processes simulation parameters and returns carbon sequestration projections over time in CSV format.

Endpoint Path: POST /rmt/simulate

Description

This endpoint accepts a simulation configuration file (typically in XML, JSON, or proprietary format) containing all parameters needed to model reforestation scenarios including species selection, planting density, management regime, site conditions, and projection timeframe. The API processes this file and returns simulation results in CSV format including carbon stock changes, biomass accumulation, and other relevant metrics over time.

Request Header

Header Name	Value	Required	Data Type	Description	Purpose
Content-Type	multipart/form-data	Yes	string	Indicates the request contains multipart form data with file upload	Enables file upload functionality
Accept	text/plain	Yes	string	Specifies the expected response format	Indicates client expects plain text response (CSV format)
Ocp-Apim-Subscription-Key	{subscription_key}	Yes	string	Azure APIM subscription key	Validates API subscription and tracks usage

Request Body (form-data)

Field Name	Data Type	Required	Format	Description
file	file	Yes	.rmd	The simulation configuration file containing all parameters for the reforestation model including site location, species, management practices, and simulation duration

Example Request

POST {{baseUrlrmt}}api/v1/rmt/simulate

Content-Type: multipart/form-data

Accept: text/plain

Ocp-Apim-Subscription-Key: YOUR_rmt_SUBSCRIPTION_KEY

--boundary

Content-Disposition: form-data; name="file"; filename="simulation_config.xml"

Content-Type: application/xml

[Binary file content]

--boundary--

Example Responses

Success Response (200 OK) - CSV Format

Year,CarbonSequestration,Biomass,AboveGroundBiomass,BelowGroundBiomass,DeadOrganicMatter,

Soil

2024,12.5,85.3,65.2,20.1,15.8,45.2

2025,25.0,120.5,92.4,28.1,18.5,48.7

2026,38.2,158.7,121.5,37.2,21.3,52.1

2027,52.1,199.3,152.6,46.7,24.8,55.9

2028,66.8,242.1,185.3,56.8,28.5,59.8

2029,82.3,287.5,220.1,67.4,32.7,64.2

2030,98.5,335.2,256.8,78.4,37.2,68.9

Bad Request Response (400)

{

"type": "https://api.dcceew.gov.au/errors/validation-error",

"title": "Validation Error",

"status": 400,

"detail": "The uploaded file is not in a valid format. Expected XML or JSON configuration file with required simulation parameters.",

"instance": "/rmt/simulate"

```
}
```

Common 400 Error Scenarios:

- Invalid file format (not XML, JSON, or recognized proprietary format)
- Missing required parameters in configuration file
- Malformed XML/JSON structure
- Invalid species ID or regime ID in configuration
- Out-of-range coordinate values
- Invalid date ranges or simulation periods
- Negative or zero values for required positive parameters
- File size exceeds maximum allowed limit

Server Error Response (500)

```
{  
  "type": "https://api.dcceew.gov.au/errors/internal-error",  
  "title": "Internal Server Error",  
  "status": 500,  
  "detail": "An unexpected error occurred while processing the simulation. The  
    simulation engine encountered an error. Please try again later or contact support.  
  "instance": "/rmt/simulate"  
  ,  
}
```

Common 500 Error Scenarios:

- Database connection failure
- Simulation engine crash or timeout
- Memory overflow during complex calculations
- External service dependency failure
- Climate data service unavailable
- Insufficient server resources

Troubleshooting

Problem: 400 Bad Request with "Invalid file format"

Solution: Verify file format matches expected schema (XML or JSON). Check file encoding (UTF-8 recommended) and structure. Validate against schema if available.

Problem: Simulation takes too long or times out

Solution: Reduce simulation complexity by shortening time horizon, reducing data frequency, or simplifying regime. Contact support for optimization guidance.

Problem: Inconsistent results between runs with same parameters

Solution: Ensure all parameters are explicitly specified in configuration file; default values may vary. Check for stochastic elements in simulation model.

Problem: CSV response is empty or malformed

Solution: Check that simulation completed successfully. Verify file upload was complete. Review simulation parameters for validity.

Problem: Unexpected carbon values (negative or extremely high)

Solution: Review input parameters, especially initial carbon stocks and management practices. Verify coordinate location is correct.

Problem: Cannot parse CSV response

Solution: Check response encoding (should be UTF-8). Verify CSV delimiter (comma). Handle potential quoted fields properly.

5. RMT Decrypt File API

Purpose: Decrypt a single encrypted RMD (Reforestation Model Data) file. This endpoint is used when simulation results or configuration files are returned in encrypted format for security purposes.

Endpoint Path: POST /rmt/decrypt-file

Description

This endpoint accepts a single encrypted RMD file and returns its decrypted contents in plain text format. RMD files are proprietary format files used by the Reforestation Modelling Tool and must have the .rmd file extension. This endpoint is useful when working with sensitive data or when simulation results are stored in encrypted format for compliance or security requirements.

Request Headers

Header Name	Value	Required	Data Type	Description	Purpose
Content-Type	multipart/form-data	Yes	string	Indicates the request contains multipart form data with file upload	Enables file upload functionality
Accept	text/plain	Yes	string	Specifies the expected response format	Indicates client expects plain text response
Ocp-Apim-Subscription-Key	{subscription_key}	Yes	string	Azure APIM subscription key	Validates API subscription and tracks usage

Request Body (Form Data)

Field Name	Data Type	Required	Format	Description
file	file	Yes	.rmd	The encrypted RMD file to be decrypted. File should be previously encrypted by the RMT system or using compatible encryption methods.

Request Body Validation Rules

1. **File Presence:** The file field must be present and contain a valid file
2. **File Extension:** File must have .rmd extension (case-insensitive)
3. **File Format:** File must be in encrypted RMD format recognized by the system
4. **File Size:** File size must not exceed maximum allowed size
5. **Encryption Method:** File must be encrypted using supported encryption algorithms
6. **File Integrity:** File must not be corrupted or tampered with

Example Requests

POST {{baseUrlrmt}}api/v1/rmt/decrypt-file

Content-Type: multipart/form-data

Accept: text/plain

Ocp-Apim-Subscription-Key: fe45f618b0e049c982374b4b8e8f9acf

--boundary

Content-Disposition: form-data; name="file"; filename="simulation_results.rmd"

Content-Type: application/octet-stream

[Binary encrypted RMD file content]

--boundary--

Example Response

Success Response (200 OK) - Decrypted Configuration File

```
<?xml version="1.0" encoding="UTF-8"?>
```

```
<ReforestationSimulation>
```

```
  <Project>
```

```
    <Name>Queensland Reforestation Project</Name>
```

```
    <Location>
```

```
      <Longitude>144.0</Longitude>
```

```
      <Latitude>-15.0</Latitude>
```

```
    </Location>
```

```
  </Project>
```

```
  <Species>
```

```
    <Id>c7db0a0134493d1019360000</Id>
```

```

    <Name>Eucalyptus camaldulensis</Name>
  </Species>

  <Regime>

    <Id>1050</Id>

    <Name>Intensive Pulpwood Production</Name>
  </Regime>

  <SimulationParameters>

    <StartYear>2024</StartYear>

    <Duration>25</Duration>

    <PlantingDensity>1250</PlantingDensity>
  </SimulationParameters>
</ReforestationSimulation>

```

Bad Request Response (400) - Decryption Failure

```

{
  "type": "https://api.dcceew.gov.au/errors/decryption-error",
  "title": "Decryption Error",
  "status": 400,
  "detail": "Unable to decrypt the provided RMD file. The file may be corrupted, not
    properly encrypted, or encrypted with an incompatible method.",
  "instance": "/rmt/decrypt-file"
}

```

Server Error Response (500)

```

{
  "type": "https://api.dcceew.gov.au/errors/internal-error",
  "title": "Internal Server Error",
  "status": 500,
  "detail": "An unexpected error occurred during decryption. The decryption service
    encountered an error. Please try again later or contact support.",
  "instance": "/rmt/decrypt-file"
}

```

Common 500 Error Scenarios:

- Decryption service unavailable
- Key management system failure
- Memory issues with large files
- Unexpected encryption format
- Database connection failure
- Timeout during decryption process

Troubleshooting

Problem: 400 Bad Request with "Invalid file extension"

Solution: Verify file has .rmd extension (case-insensitive). Rename file if necessary. Do not simply rename non-RMD files.

Problem: 400 Bad Request with "Unsupported encryption method"

Solution: Verify files were encrypted using RMT-compatible encryption. Contact support for encryption specifications. Do not use third-party encryption tools.

Problem: Decryption succeeds but content is garbled or unreadable

Solution: Check file encoding (should be UTF-8). Ensure complete file upload. Verify file wasn't corrupted during transfer. Try re-downloading original encrypted file.

Problem: Large files fail to decrypt

Solution: Check file size against API limits. Split large files if possible. Contact support for increased limits if needed.

Problem: Intermittent decryption failures

Solution: Check network stability during upload. Implement retry logic with exponential backoff. Verify file integrity before upload using checksums.

Problem: Decrypted content doesn't match expected format

Solution: Verify you're decrypting the correct file type. Check original file source. RMD files can contain various data types.

Problem: 500 Internal Server Error during decryption

Solution: Wait and retry with exponential backoff. If problem persists, contact support with file details (but not the file itself).

Support and Contact

- **Email:** fullcam@dcceew.gov.au
- **Terms of Service:** [FullCAM Terms](#)
- **License Agreement:** [FullCAM License](#)

When Contacting Support for Batch processing related issues

Please provide:

1. Batch ID — The identifier for the affected batch
2. Trace ID — From any error responses received
3. Timestamps — When the issue occurred (UTC preferred)
4. Steps to reproduce — What you were doing when the issue occurred
5. Expected vs actual behaviour — What you expected result and what actually result

API Usage Guide:

For detailed instructions on obtaining subscription keys and getting started with the API, refer to the FullCAM Modernisation API Usage Guide available at:

[FullCAM_Modernisation_API_Usage_Guide.pdf](#)

Glossary

Term	Definition
API	Application Programming Interface
REST API	Representational State Transfer Application Programming Interface
FullCAM PR	Full Carbon Accounting Model Public Release
FullCAM PR version	Full Carbon Accounting Model Public Release Version (2020, 2024)
API Authentication	An API authentication is the process of verifying the identity of a user or application before granting access to an API
Base URL	An API Base URL is the primary or starting URL for an API, acting as a common prefix for all subsequent API Endpoints
API Endpoint	An API Endpoint is a specific location (typically a URL) on a server where an API (Application Programming Interface) receives requests for resources or functionality
API Request	An API request, also known as an API call, is a message sent from a client application to a server to access a specific resource or perform an action via an API (Application Programming Interface)
API Response	An API response is the data returned by a server in response to a client's API request
RMD File	Reforestation Model Data file. Proprietary format file used by the RMT system, typically with .rmd extension.
Plot File	FullCAM Plot file, Proprietary format file used by the FullCAM system, typically with .plo extension.
PLD	FullCAM Plot Diest file Proprietary format file used by the FullCAM system, typically with .pld extension.
EST	FullCAM Estate file Proprietary format file used by the FullCAM system, typically with .pld extension.